

# Summary of the interviews

2025-04-08

## Treatment

## Methodology

Score is assigned to each answer according to following flowchart:

- if there is a number take it
- assign codes 999, -999, -998

For those -999:

- count and exclude

For those 999:

- if there is an obvious keyword use it for cross scoring
- if there is no obvious keyword and the question is needed for the CPT use LLM
- if there is no obvious keyword and the question is needed for the DAG review by expert
- any other: review by expert

For those -998: - try to assign a score borrowing from other people

## Results

### Cross-scoring

### LLM scoring

### The IDK

### Missing interpretation

## Interviews Text

Legend:

-999: does not apply -998: applies but did not know -997: the reply does not match the question 999: pending interpretation 0-5

## Interviewee: I 1

1. name: peter berggren. category: fishing guide id1 is a fishing guide focused on the waters around stockholm, both in the archipelago and in lakes. he guides mainly for trout, pike, and perch, and makes a living from his business. on average, they guide about 100 days per year. the guiding business is seasonal and follows the fish's behaviour, activity, and migrations. they fish in lakes, streams and coastal areas. they can use boats in coastal areas and lakes. it is more common in spring and summer. in winter they guide ice fishing. if there is no ice they can switch to spinning but this is unpredictable. the most fished species are sea trout, salmon, pike, and perch. they primarily practice catch and release fishing. because fishing guide is mostly a touristic activity, it is more active in the weekend, and their flexibility is linked to customers flexibility. how they adapt: fishing guide adapts all the time according to the conditions and the clients need. they are flexible for place, species and gear, but its ability to adapt to conditions depends on client's flexibility. he notices a decline in demand during extreme hot weather, and for security reason he cannot guide if he feels danger. in winter he can replace ice fishing with spinning but cannot book this in advance. in summer he can choose between different options depending on the conditions. he can change fish location and target (and accordingly gear) in heatwaves, algal bloom and stormy conditions. in hw he can move the trips to mornings and evenings. in hw and ab he moves from shallow waters for pike to below the thermocline for sea trout; in stormy conditions he goes the other way around.

## Q 2 - go\_out

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1–5; from 1 – very unlikely, to 5 – very likely]

|   | Event                                     | Answer                                                                                                                                                                                                                                                                                                                                                                                  | Score | Method |
|---|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | Very different, depends on what is booked. Sometimes ice fishing is booked and then we have to cancel all bookings, sometimes open water fishing is possible. Lose ice fishing customers and gain spin fishing ones; 1 ice fishing (spin fishing customers can't be booked in advance in winter, you can't count on a heatwave in winter)                                               | NA    | M      |
| b | Heatwave in the summer season             | 4. Some fishing is closed if it gets too hot, e.g., river fishing. It's difficult to fish if it gets extremely hot, but I try to adapt; so far I haven't canceled coastal fishing, I move it to mornings & evenings. But, for example, Nyköpingsån was closed because the water was too warm. Also depends on how hot it is. It happened in 2018, depends on what happens with our fish | 4     | O      |
| c | Severe icing (in the winter season)       | 1. Haven't experienced severe icing, only seen it on YouTube; don't know if it happens on sportfishing boats; it has to be safe, not dangerous; if it's icing like on the Great Lakes, then the boat breaks                                                                                                                                                                             | 1     | O      |
| d | Algal bloom (in the summer season)        | 1. Usually happens offshore; can't handle it if I have a trip, e.g., from Sandhamn, but then we can go inland when there's algal bloom                                                                                                                                                                                                                                                  | 1     | O      |
| e | Extreme cloudburst (in the summer season) | 1, dangerous                                                                                                                                                                                                                                                                                                                                                                            | 1     | O      |
| f | Gale                                      | 5Gale is normal (sometimes we cancel), from strong gale and upwards we stay home (18 19 m/s average and 25 m/s in gusts is not possible)                                                                                                                                                                                                                                                | 5     | O      |
| g | Storm                                     | 1                                                                                                                                                                                                                                                                                                                                                                                       | 1     | O      |

|   |                                                                                              |                                                                                                                                                                                               |    |    |
|---|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|
| h | Are there any of these in combination that affect how likely it is that you go out and fish? | A lot is about safety, combinations become dangerous, e.g., strong icing and strong winds, I avoid this as much as possible; would also get fewer requests, since fishing should be enjoyable | NA | NA |
|---|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|

Multiple options identified

|   | Event                         | gear        | Score (given) |
|---|-------------------------------|-------------|---------------|
| a | Heatwave in the winter season | ice fishing | 1             |
| a | Heatwave in the winter season | rod         | 5             |

### Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better]

|   | Event                                     | Answer                                                                                                                                                                                                          | Score | Method |
|---|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | 1 ice fishing, there is no catchability if there are no fish; however, the fish like when it gets warmer, so if it's spin fishing then 5                                                                        | NA    | M      |
| b | Heatwave in the summer season             | 2                                                                                                                                                                                                               | 2     | O      |
| c | Severe icing (in the winter season)       | No, can't conduct fishing; if it is possible to fish then it becomes 2.                                                                                                                                         | 2     | O      |
| d | Algal bloom (in the summer season)        | 2, the fish move from that area; also becomes difficult fishing (fished during an algal bloom this summer, had to go all the way into the bay because where we intended to fish we only got weeds on the hook). | 2     | O      |
| e | Extreme cloudburst (in the summer season) | Don't go out in cloudbursts; fish react to pressure changes; if there is low pressure during a downpour, then they stop biting, but if it rises after the downpour then it can be good; so I can't say          | -998  | NA     |
| f | Gale                                      | : 2, a gale doesn't matter for the fishing itself, other than that the gear is hard to use                                                                                                                      | 2     | O      |
| g | Storm                                     | Don't go out in a storm                                                                                                                                                                                         | -998  | NA     |

Multiple options identified

|   | Event                         | gear        | Score (given) |
|---|-------------------------------|-------------|---------------|
| a | Heatwave in the winter season | ice fishing | 1             |
| a | Heatwave in the winter season | rod         | 5             |

#### Q 4 - catch\_assemblage

Do you catch a different composition of species during such extreme weather events?

[1–5; from 1 – very similar, to 5 – very different species composition]

|   | Event                                     | Answer                                                                                                                               | Score | Method |
|---|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | : 5 sea trout on open water, instead of perch and pike under the ice                                                                 | 5     | O      |
| b | Heatwave in the summer season             | : 5 redirect the fishing down below the thermocline                                                                                  | 5     | O      |
| c | Severe icing (in the winter season)       | No information                                                                                                                       | -998  | NA     |
| d | Algal bloom (in the summer season)        | More about changing location (deeper), 5 choose other species (we fish pike & perch, sea trout)                                      | 5     | O      |
| e | Extreme cloudburst (in the summer season) | No other species, more that we don't fish                                                                                            | -998  | NA     |
| f | Gale                                      | : 2 sometimes; if it blows too hard and we are supposed to fish trout, then sometimes we have to go in and fish perch & pike instead | 2     | O      |
| g | Storm                                     | Cancel                                                                                                                               | -998  | NA     |

#### Q 5 - catch\_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

|   | Event                                     | Answer          | Score | Method |
|---|-------------------------------------------|-----------------|-------|--------|
| a | Heatwave in the winter season             | 3               | 3     | O      |
| b | Heatwave in the summer season             | 2, a b it worse | 2     | O      |
| c | Severe icing (in the winter season)       | 3               | 3     | O      |
| d | Algal bloom (in the summer season)        | 2               | 2     | O      |
| e | Extreme cloudburst (in the summer season) | 3               | 3     | O      |
| f | Gale                                      | 3               | 3     | O      |
| g | Storm                                     | 3               | 3     | O      |

### Q 6 - cope\_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1–5; from 1 – very unlikely, to 5 – very likely]

We release 98% of the fish; we are more concerned with the survival of the fish, e.g., during a heatwave survival becomes difficult; we don't want to fish pike when it's 30 degrees, then we'd rather fish sea trout that can survive better when we release it, since it goes down to colder depth

|   | Event                                     | Answer | Score | Method |
|---|-------------------------------------------|--------|-------|--------|
| a | Heatwave in the winter season             | NA     | -999  | NA     |
| b | Heatwave in the summer season             | 3      | 3     | O      |
| c | Severe icing (in the winter season)       | NA     | -999  | NA     |
| d | Algal bloom (in the summer season)        | 3      | 3     | O      |
| e | Extreme cloudburst (in the summer season) | NA     | -999  | NA     |
| f | Gale                                      | NA     | -999  | NA     |
| g | Storm                                     | NA     | -999  | NA     |

### Q 7 - lag\_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

4. Yes, I do have to change the setup of the fishing. With milder winters, I can't fish or might even take a regular job

|   | Event                                     | Answer | Score | Method |
|---|-------------------------------------------|--------|-------|--------|
| a | Heatwave in the winter season             | NA     | 4     | NA     |
| b | Heatwave in the summer season             | NA     | -999  | NA     |
| c | Severe icing (in the winter season)       | NA     | -999  | NA     |
| d | Algal bloom (in the summer season)        | NA     | -999  | NA     |
| e | Extreme cloudburst (in the summer season) | NA     | -999  | NA     |

|   |       |    |      |    |
|---|-------|----|------|----|
| f | Gale  | NA | -999 | NA |
| g | Storm | NA | -999 | NA |

### Q 8 - unc\_2\_7

How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? 1–5; from 1 – not at all, to 5 – very much

| Answer                                                            | Score | Method |
|-------------------------------------------------------------------|-------|--------|
| 5, since I am chairman I have the mandate to represent the others | - 997 | NA     |

### Q 9 - personal\_safety

How much concern for your personal safety do these extreme weather events bring?

[0–2; 0 – no concern, 1 – personal injuries, 2 – life-threatening

|   | Event                                     | Answer                                                             | Score | Method |
|---|-------------------------------------------|--------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | 1 I do it for ice fishing; if no one falls in now, it is a miracle | 1     | O      |
| b | Heatwave in the summer season             | 0                                                                  | 0     | O      |
| c | Severe icing (in the winter season)       | 2                                                                  | 2     | O      |
| d | Algal bloom (in the summer season)        | 0                                                                  | 0     | O      |
| e | Extreme cloudburst (in the summer season) | 1                                                                  | 1     | O      |
| f | Gale                                      | 1 (ordinary gale none, but strong gale)                            | 1     | O      |
| g | Storm                                     | 2                                                                  | 2     | O      |

### Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

|   | Event                         | Answer | Score | Method |
|---|-------------------------------|--------|-------|--------|
| a | Heatwave in the winter season | 0      | 0     | O      |
| b | Heatwave in the summer season | 0      | 0     | O      |

|   |                                           |                                                                                                                                                                                                                    |   |   |
|---|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
| c | Severe icing (in the winter season)       | don't know, probably 2. (have broken parts on the boat twice because of it)                                                                                                                                        | 2 | O |
| d | Algal bloom (in the summer season)        | 0                                                                                                                                                                                                                  | 0 | O |
| e | Extreme cloudburst (in the summer season) | don't know, more if the boat sinks; is that a 3 then, don't know, depends on how big the damage is; in combination with storm then it becomes 3. Usually, you get away from a cloudburst. There is a risk; maybe 2 | 2 | O |
| f | Gale                                      | 1                                                                                                                                                                                                                  | 1 | O |
| g | Storm                                     | 3                                                                                                                                                                                                                  | 2 | G |

### Q 11 - minor\_damage\_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

| Answer    | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit     |
|-----------|---------------------------|--------------------------|----------|
| 10 20k kr | 10000                     | 20000                    | sek_unit |

### Q 12 - major\_damage\_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

| Answer     | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit     |
|------------|---------------------------|--------------------------|----------|
| 20 600k kr | 20000                     | 6e+06                    | sek_unit |

### Q 13 - change\_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0-2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

Change all the time according to the conditions; place, species and gear

|   | Event                               | Answer | Score | Method |
|---|-------------------------------------|--------|-------|--------|
| a | Heatwave in the winter season       | NA     | -999  | NA     |
| b | Heatwave in the summer season       | NA     | -999  | NA     |
| c | Severe icing (in the winter season) | NA     | -999  | NA     |

|   |                                           |    |      |    |
|---|-------------------------------------------|----|------|----|
| d | Algal bloom (in the summer season)        | NA | -999 | NA |
| e | Extreme cloudburst (in the summer season) | NA | -999 | NA |
| f | Gale                                      | NA | -999 | NA |
| g | Storm                                     | NA | -999 | NA |

#### Q 14 - unc\_9\_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

| Answer                                   | Score | Method |
|------------------------------------------|-------|--------|
| great similarity between boat guides (1) | 1     | NA     |

#### Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

Easy for me to switch but ask what the customers want (e.g., Americans who are booked for ice fishing). Large differences from the normal are a problem, with bookings we do not know what we shall book (e.g., closed Nyköpingsån fishing due to heatwave). The customers also have to be flexible. 2 for those that are not dangerous; 1 for what is dangerous (cannot conduct activities)

|   | Event                                     | Answer | Score | Method |
|---|-------------------------------------------|--------|-------|--------|
| a | Heatwave in the winter season             | NA     | 1     | NA     |
| b | Heatwave in the summer season             | NA     | 2     | NA     |
| c | Severe icing (in the winter season)       | NA     | 1     | NA     |
| d | Algal bloom (in the summer season)        | NA     | 2     | NA     |
| e | Extreme cloudburst (in the summer season) | NA     | 1     | NA     |
| f | Gale                                      | NA     | 1     | NA     |
| g | Storm                                     | NA     | 1     | NA     |

#### Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?



[0–2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

| Answer                                                                                                                                                                                                                                                                                                                                                    | Score | Method |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| 0 , if you take the extreme weather event to its peak, extreme cloudburst or storm, then the boat does not help; same if it is extreme heatwave, then we cannot go out 38 42 °C; in 2018 we could not book the fishing. Another parameter: our activity also depends on demand, and demand is made impossible during many of these extreme weather events | 0     | O      |

### Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer | Score | Method |
|--------|-------|--------|
| 2      | 2     | O      |

### Q 18 - management\_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer                                                           | Score | Method |
|------------------------------------------------------------------|-------|--------|
| Sport fishing has great mobility; the accessibility to water. 2. | 2     | O      |

### Q 19 - management\_species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer | Score | Method |
|--------|-------|--------|
| 2 Yes  | 2     | O      |

### Q 20 - unc\_15\_19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

| Answer | Score | Method |
|--------|-------|--------|
|--------|-------|--------|

|                                                                                                                                                                                                             |   |    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|
| Probably varies, yes it probably varies between fishers: some have a home harbor and are not so flexible; does that apply to management too do you think? don't know, generally not so much variation there | 2 | NA |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|

### Q 21 - stress

How much stress does the risk of the following events cause (in general)?

[1–5; from 1 – really not stressful, to 5 – very stressful]

|   | Type             | Answer                                                                                                           | Score | Method |
|---|------------------|------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | stress_injuries  | 5                                                                                                                | 5     | O      |
| b | stress_gear      | 5                                                                                                                | 5     | O      |
| c | stress_condition | 4 (one does not want to fish if the fish are not doing well, it is also bad promotion for me as a fishing guide) | 4     | O      |

### Q 22 - unc\_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

| Answer                                                                                                                                                                  | Score | Method |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| it is so personal, some do not care at all, they just drink beer all day; we discuss things like this all the time, most are probably quite stressed by such situations | 4     | NA     |

### Q 23 - job\_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

| Answer                                                                                                                                                                                                  | Score | Method |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| I have to work, so I have no choice; everyone has the opportunity to work with something else who has two legs and two arms; I only guide, I have nothing I can lean back on. Is it 3? I would have to. | 3     | O      |

### Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

| Answer                                                                                                                                                                                                                     | Score | Method |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| Would probably try to find solutions; sport fishing itself would certainly change, and then fishing guides would follow that change; there is probably some variation among guides depending on what one is specialized in | 4     | NA     |

**Q 25 - credit**

How easy is it to gain access to credits, financing or loans for investments

[1–5; from 1 – very difficult, to 5 – very easy]

| Answer                                                                                                                                                                               | Score Method |   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|
| 1. there are no possibilities; completely self financed. The bank would just laugh at one. The businesses are too small, and no one invests in this. Our whole industry is like that | 1            | O |

**Q 26 - unc\_23\_25**

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

| Answer                                                                                   | Score Method |    |
|------------------------------------------------------------------------------------------|--------------|----|
| see answer above; the question was not asked, because the answers had already been made. | 2            | NA |

**Q 27 - knowledge\_transmission**

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Score Method |   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|
| the world does not end without fishing guides; but we are ambassadors for how the development of fishing continues; it matters a lot how we handle fish & fishing in front of the general sport fishing mass, which is very large, 20% of Sweden's population; 5. important for a positive handling of fish and restoration of habitat. We are the only voice for the fish in the water. Important that there is a small scale tourist fishing industry, for a positive development for the fish | 5            | O |

**Q 28 - benefit\_local\_community**

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                                                                                   | Score Method |   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|
| quite a lot; 3; influences so that more sport fishing equipment is sold in the nearby shops, hotel nights, everything around lunches, dinners, and so on | 3            | O |

**Q 29 - benefit\_personal\_community**

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

| Answer               | Score Method |   |
|----------------------|--------------|---|
| 5. see answer to 27. | 5            | O |

### Q 30 - traditional\_food

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                            | Score Method |   |
|-------------------------------------------------------------------|--------------|---|
| 1. fish are not cows. We do not fish for food, almost none of us. | 1            | O |

### Q 31 - unc\_26\_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

| Answer             | Score Method |   |
|--------------------|--------------|---|
| great variation 5. | 5            | O |

### Q 32 - outdoor\_benefit

How important is it for you to be outdoors (when you fish) for your well-being

[1–5; from 1 – not important at all, to 5 – very important]

| Answer | Score Method |   |
|--------|--------------|---|
| 5      | 5            | O |

### Q 33 - identity

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

| Answer | Score Method |   |
|--------|--------------|---|
| 5      | 5            | O |

### Q 34 - home

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

| Answer                                                                            | Score Method |   |
|-----------------------------------------------------------------------------------|--------------|---|
| 5, it creates positive memories; that is how one creates a relationship to places | 5            | O |

**Q 35 - unc\_31\_34**

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

| Answer                                                                                 | Score Method |   |
|----------------------------------------------------------------------------------------|--------------|---|
| it is probably very unanimous among sport fishers; that is why one works with this; 1. | 1            | O |

**Q 36 - fdays**

Approximately how many days per year do you fish?

[free text description]

| Answer | Score Method |   |
|--------|--------------|---|
| 100    | 100          | O |

**Q 37 - income**

Approximately how large is your gross income from a fishing day on average?

[free text description]

| Answer                                                      | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit    |
|-------------------------------------------------------------|---------------------------|--------------------------|---------|
| 6000 8000 SEK/day and boat, generally the same for everyone | 6000                      | 8000                     | sek_day |

**Q 38 - season\_maint**

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

| Answer                                                                                                                                                                                                                                                                                                                                                    | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|------------|
| oh, you don't want to know! what I do is so uneconomical; if the engine breaks then several 100,000 SEK go up in smoke, electric motor 50,000 SEK, sonar 50,000 SEK; is it rather 100,000 200,000 than 10,000 50,000 SEK? yes. Large part is expenses for material, applies also to the cars; 5 days a week fishing, 1 service day per week, 1 family day | 50000                     | 10000                    | sek_season |

**Q 39 - fuel\_trip**

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

| <b>Answer</b>                                                                  | <b>Score<br/>- low<br/>(given)</b> | <b>Score<br/>- up<br/>(given)</b> | <b>Unit</b> |
|--------------------------------------------------------------------------------|------------------------------------|-----------------------------------|-------------|
| maybe 1000 SEK/fishing day; this is the biggest expense besides food for a day | 1000                               | 1000                              | sek_day     |

#### Q 40 - extra\_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

| <b>Answer</b>                         | <b>Score<br/>- low<br/>(given)</b> | <b>Score<br/>- up<br/>(given)</b> | <b>Unit</b> |
|---------------------------------------|------------------------------------|-----------------------------------|-------------|
| 1500 SEK for such a longer travel day | 500                                | 500                               | sek_day     |

#### Q 41 - extra\_crew

What are the costs for extra crew on board?

[free text description]

| <b>Answer</b>                                                                                                                                                                                                  | <b>Score<br/>- low<br/>(given)</b> | <b>Score<br/>- up<br/>(given)</b> | <b>Unit</b> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-----------------------------------|-------------|
| I always guide myself; there are some who sometimes take in an extra person; quite expensive, since it is salary and employer's contributions etc. (guess they get 1500 SEK, so in total it becomes 3000 SEK). | 2000                               | 3000                              | sek_day     |

#### Q 42 - extra\_cleaning\_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

| <b>Answer</b>                                                                                                                                                                                                                                                  | <b>Score<br/>- low<br/>(given)</b> | <b>Score<br/>- up<br/>(given)</b> | <b>Unit</b> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-----------------------------------|-------------|
| a whole workday for service, so it costs what I earn on a fishing day; no, not for algal bloom, but in gale or storm there are many lines that must be fixed; twice I have broken the boat due to icing, the trim motor and the steering, 15,000 SEK per time. | NA                                 | NA                                | sek_day     |

#### Q 43 - extra\_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?

[free text description]

|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|--|--|--|--|

| Answer                                                                                                                                                                                                                                                                                                                                                                                                                         | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|---------|
| the biggest cost is the loss of income. Cancelled fishing days. A tourist usually does not rebook, especially if it is short notice change then I don't have time to get in something else. Guides for corporate events are at 30,000 SEK, or 60,000 SEK when we are out with 6 boats; if you miss that then the whole income disappears; loss of income also arises because people do not book when there is extreme weather. | NA                        | NA                       | sek_day |

#### Q 44 - unc\_36\_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

| Answer                    | Score | Method |
|---------------------------|-------|--------|
| significant variation; 5. | 5     | NA     |

#### Q 45 - NA

Anything you want to add?

[free text description]

| Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Score     | Method     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------|
| There may be disease among the fish during heatwaves, and that can have consequences for fishing for five years ahead; and algal bloom we do not know, but it does not feel good. We are completely dependent on the fish being well. How have perch and pike been doing this summer?<br>It was a bit weak when it was hot, but now they have started again. Probably quite a normal year. One adapts to the new. Sour that one cannot fish in the outer archipelago. It is an enormous difference from how it was when I was little, we would need supportive stocking of the pike stocks. First year with protection. I have not seen so much herring in 15 years! We have had incredibly little fish coming in to spawn earlier; when they last did test fishing at Mörkö, there was an 84 86% decline in spawning herring. Out last week, then there was herring in Furusundsleden, a lot of herring, proper schools. Summer herring. Don't know when I last saw that. Very positive! Will have great impact on the predatory fish stocks. Hope that the seals can move out again, so they don't have to hang around | t<br>Slus | en.<br> -9 |

#### Interviewee: I 2

- name: teija aho. category: coastal fisherman id2 is a professional costal fisherman who operates in the north of bothnian bay (kalix). they fish in coastal waters with gillnets and traps (salmon traps and other traps for perch and bream), and they operate from boat at very short distances from their harbour. the target they mention is perch, bream, herring, salmon. they also run vendace fishery for roe production. the main fishing season is summer (from april to august). in winter they generally do not go out until spring comes. they can do sporadically some ice fishing. how they adapt: id2 fish has a structured succession of fishing styles within season. depending on the moment, they want to aim for something and may be forced to change. in the case they are fishing with nets (for pike, perch and herring?) they are flexible. their options to avoid hw and ab are to fish deeper, to avoid storms is to

choose a different bay. hw especially affects perch, which die immediately. they notice more breams with hw. salmon traps are very rigidly managed and tied to a unique boat. salmon traps do not suffer from hw, but they do of storm, so if they must haul them up with a bigger boat, they get a penalty. vendace season is also rigid and short.

## Q 2 - go\_out

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1–5; from 1 – very unlikely, to 5 – very likely]

|   | Event                                     | Answer                                                                                      | Score | Method |
|---|-------------------------------------------|---------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | Extremely warm winter we ice fish so little, therefore it is a bit difficult to answer. (2) | -999  | G      |
| b | Heatwave in the summer season             | (3) Depends a little on what we are fishing                                                 | 3     | O      |
| c | Severe icing (in the winter season)       | 1                                                                                           | 1     | O      |
| d | Algal bloom (in the summer season)        | Doesn't affect us much up here; locally there may be algal bloom in a bay. (4)              | 4     | O      |
| e | Extreme cloudburst (in the summer season) | 3                                                                                           | 3     | O      |
| f | Gale                                      | 1, not me at least with my small boat                                                       | 1     | O      |
| g | Storm                                     | 1                                                                                           | 1     | O      |

## Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better]

|   | Event                                     | Answer                                                                                                                               | Score | Method |
|---|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | 3                                                                                                                                    | -999  | G      |
| b | Heatwave in the summer season             | 2, Becomes worse, maybe not for all species, but they move to other areas, so you have to search elsewhere                           | 2     | O      |
| c | Severe icing (in the winter season)       | Hypothetical question, since we always have ice in winter; icing can happen in autumn, but we rarely fish then, so can't answer that | -998  | NA     |
| d | Algal bloom (in the summer season)        | Hard to answer, doesn't occur very often                                                                                             | -998  | NA     |
| e | Extreme cloudburst (in the summer season) | 3                                                                                                                                    | 3     | O      |
| f | Gale                                      | It becomes harder to fish, but does not affect the actual catch; (2). Can affect a little since it's harder to fish                  | 2     | O      |



|   |       |                                                    |   |    |
|---|-------|----------------------------------------------------|---|----|
| g | Storm | In a storm you can't fish, so catchability is zero | 1 | NA |
|---|-------|----------------------------------------------------|---|----|

#### Q 4 - catch\_assemblage

Do you catch a different composition of species during such extreme weather events?

[1–5; from 1 – very similar, to 5 – very different species composition]

|   | Event                                     | Answer                                                                                                | Score | Method |
|---|-------------------------------------------|-------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | 1                                                                                                     | -999  | G      |
| b | Heatwave in the summer season             | Somewhat different species if we fish in the same place e.g., a lot of bream; somewhat different (4). | 4     | O      |
| c | Severe icing (in the winter season)       | NA                                                                                                    | -998  | NA     |
| d | Algal bloom (in the summer season)        | NA                                                                                                    | -998  | NA     |
| e | Extreme cloudburst (in the summer season) | 1                                                                                                     | 1     | O      |
| f | Gale                                      | 1                                                                                                     | 1     | O      |
| g | Storm                                     | NA                                                                                                    | -998  | NA     |

#### Q 5 - catch\_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

|   | Event                                     | Answer                                                                                                                                                                                                             | Score | Method |
|---|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | 3                                                                                                                                                                                                                  | -999  | G      |
| b | Heatwave in the summer season             | 1 Yes, then it gets quite a lot worse. Perch cannot tolerate lying in nets; they die almost immediately. Salmon in traps is a short fishing period and in deep water, so it doesn't get warm enough to affect them | NA    | M      |
| c | Severe icing (in the winter season)       | NA                                                                                                                                                                                                                 | -999  | NA     |
| d | Algal bloom (in the summer season)        | If there were to be strong algal bloom, which hasn't happened so far, then it would certainly affect negatively hypothetical.                                                                                      | 2     | NA     |
| e | Extreme cloudburst (in the summer season) | NA                                                                                                                                                                                                                 | -998  | NA     |
| f | Gale                                      | NA                                                                                                                                                                                                                 | -998  | NA     |

|   |       |    |      |    |
|---|-------|----|------|----|
| g | Storm | NA | -998 | NA |
|---|-------|----|------|----|

Multiple options identified

|   | Event                         | target | Score (given) |
|---|-------------------------------|--------|---------------|
| b | Heatwave in the summer season | perch  | 1             |
| b | Heatwave in the summer season | salmon | 3             |

### Q 6 - cope\_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1–5; from 1 – very unlikely, to 5 – very likely]

|   | Event                                     | Answer                                           | Score | Method |
|---|-------------------------------------------|--------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | NA                                               | -999  | NA     |
| b | Heatwave in the summer season             | Fish in deeper waters, maybe a middle value (3). | 3     | O      |
| c | Severe icing (in the winter season)       | NA                                               | -999  | NA     |
| d | Algal bloom (in the summer season)        | Fish in deeper waters, not difficult to avoid    | 4     | NA     |
| e | Extreme cloudburst (in the summer season) | NA                                               | -999  | NA     |
| f | Gale                                      | NA                                               | -999  | NA     |
| g | Storm                                     | NA                                               | -999  | NA     |

### Q 7 - lag\_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

1–5; from 1 – not at all, to 5 – very much:

(3) Wind and heat will have long term impacts

|   | Event                         | Answer | Score | Method |
|---|-------------------------------|--------|-------|--------|
| a | Heatwave in the winter season | NA     | 3     | NA     |

|   |                                           |    |      |    |
|---|-------------------------------------------|----|------|----|
| b | Heatwave in the summer season             | NA | 3    | NA |
| c | Severe icing (in the winter season)       | NA | -998 | NA |
| d | Algal bloom (in the summer season)        | NA | -998 | NA |
| e | Extreme cloudburst (in the summer season) | NA | -998 | NA |
| f | Gale                                      | NA | 3    | NA |
| g | Storm                                     | NA | 3    | NA |

#### Q 8 - unc\_2\_7

How much do you think the answers to these questions (questions 2-7) vary between fishers in the group you represent? 1-5; from 1 – not at all, to 5 – very much

| Answer                                                       | Score | Method |
|--------------------------------------------------------------|-------|--------|
| 1 Very little the conditions are about the same for everyone | 1     | O      |

#### Q 9 - personal\_safety

How much concern for your personal safety do these extreme weather events bring?

[0-2; 0 – no concern, 1 – personal injuries, 2 – life-threatening

|   | Event                                     | Answer             | Score | Method |
|---|-------------------------------------------|--------------------|-------|--------|
| a | Heatwave in the winter season             | 0                  | 0     | O      |
| b | Heatwave in the summer season             | 0                  | 0     | O      |
| c | Severe icing (in the winter season)       | 2 boat may sink    | 2     | O      |
| d | Algal bloom (in the summer season)        | 0                  | 0     | O      |
| e | Extreme cloudburst (in the summer season) | 0                  | 0     | O      |
| f | Gale                                      | 0 no major concern | 0     | O      |
| g | Storm                                     | 2                  | 2     | O      |

#### Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

|   | Event                                     | Answer                                                                                           | Score | Method |
|---|-------------------------------------------|--------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | 0) No boat used, the snowmobile can get stuck, but no damage, only extra work                    | -999  | G      |
| b | Heatwave in the summer season             | More work cleaning gear from fouling, (1) if that can even be called damage.                     | 1     | O      |
| c | Severe icing (in the winter season)       | 1                                                                                                | 1     | O      |
| d | Algal bloom (in the summer season)        | 0                                                                                                | 0     | O      |
| e | Extreme cloudburst (in the summer season) | (1) If the boat sinks or the engine ends up under water (that's enough for one summer).          | 1     | O      |
| f | Gale                                      | 1                                                                                                | 1     | O      |
| g | Storm                                     | (2) Mostly to gear. It very rarely happens that we lose gear completely, but a small risk exists | 2     | O      |

#### Q 11 - minor\_damage\_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

| Answer                                  | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit     |
|-----------------------------------------|---------------------------|--------------------------|----------|
| Tens of thousands of SEK, nets and such | 10000                     | 50000                    | sek_unit |

#### Q 12 - major\_damage\_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

| Answer                       | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit     |
|------------------------------|---------------------------|--------------------------|----------|
| Several hundred thousand SEK | 1e+05                     | 5e+05                    | sek_unit |

#### Q 13 - change\_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0–2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

|   | Event                         | Answer                                                   | Score | Method |
|---|-------------------------------|----------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season | (1) We might choose areas where travel on ice is easier. | 1     | O      |

|   |                                           |                                                                                            |      |    |
|---|-------------------------------------------|--------------------------------------------------------------------------------------------|------|----|
| b | Heatwave in the summer season             | (1) Might change areas a little bit                                                        | 1    | O  |
| c | Severe icing (in the winter season)       | We fish so little during that season that it's hard to say                                 | -999 | NA |
| d | Algal bloom (in the summer season)        | 0                                                                                          | 0    | O  |
| e | Extreme cloudburst (in the summer season) | (0) We wait until the downpour is over and then go out; usually it only lasts a few hours. | 0    | O  |
| f | Gale                                      | (2) Then we may not go out at all.                                                         | 2    | O  |
| g | Storm                                     | (2) Then it's quite certain we won't go out.                                               | 2    | O  |

#### Q 14 - unc\_9\_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

| Answer | Score | Method |
|--------|-------|--------|
| 1      | 1     | O      |

#### Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

|   | Event                                     | Answer                                                                                                 | Score | Method |
|---|-------------------------------------------|--------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | Not so easy, rather difficult (2).                                                                     | 2     | O      |
| b | Heatwave in the summer season             | (2) Quite difficult.                                                                                   | 2     | O      |
| c | Severe icing (in the winter season)       | can't answer                                                                                           | -999  | NA     |
| d | Algal bloom (in the summer season)        | Quite easy (4).                                                                                        | 4     | O      |
| e | Extreme cloudburst (in the summer season) | Doesn't affect much, so (not scored).                                                                  | -999  | NA     |
| f | Gale                                      | When it's windy, it's windy everywhere, so quite difficult (1) although you can always find some spot. | 1     | O      |
| g | Storm                                     | Impossible to fish at all (1).                                                                         | 1     | O      |

#### Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

| Answer | Score Method |   |
|--------|--------------|---|
| 1      | 1            | O |

### Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer                                                                            | Score Method |   |
|-----------------------------------------------------------------------------------|--------------|---|
| We only have one landing harbor, so it requires us to stay in the local area. (0) | 0            | O |

### Q 18 - management\_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                | Score Method |    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----|
| Generally, we have quite a lot of mobility in terms of management; fixed gear locations are accessible fishing spots. Only for salmon fishing the license is tied to a specific boat, so you can't switch to a larger boat. For net fishing it doesn't matter. (Example given: switching to a larger and safer boat than the one tied to the salmon license in order to haul the salmon trap during gale, but then getting a penalty on the license). | 999          | NA |

### Q 19 - management\_species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer                                                                                                                                                                                                                                    | Score Method |   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|
| 1 Depends on the fishing method. Salmon is the most restricted by management; it's the hardest to switch between species. Vendace (siklõja) too, because of the short season. At that time, we have no other alternative species to fish. | 1            | O |

### Q 20 - unc\_15\_19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

| Answer              | Score | Method |
|---------------------|-------|--------|
| 1 same for everyone | 1     | O      |

### Q 21 - stress

How much stress does the risk of the following events cause (in general)?

[1–5; from 1 – really not stressful, to 5 – very stressful]

|   | Type             | Answer                                                                          | Score | Method |
|---|------------------|---------------------------------------------------------------------------------|-------|--------|
| a | stress_injuries  | Personal injury during fishing: 3                                               | 3     | O      |
| b | stress_gear      | Losing fishing gear: 2 not so stressful since it is quite rare                  | 2     | O      |
| c | stress_condition | Low quality of the fish in the catch: 2 possible to avoid so not very stressful | 2     | O      |

### Q 22 - unc\_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

| Answer                                                                      | Score | Method |
|-----------------------------------------------------------------------------|-------|--------|
| 4 It varies a lot; some don't dare to go out when it's just a little windy. | 5     | G      |

### Q 23 - job\_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

| Answer                                                                                                                                                                                                                                           | Score | Method |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| Personally: (5). For Bothnian Bay fishers in general: If a large part of the vendace season were stormy, it would be very hard to replace it with other income. Most have other jobs in winter, but fishing is such a significant part of income | 1     | G      |

### Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

| Answer                                                                                                      | Score | Method |
|-------------------------------------------------------------------------------------------------------------|-------|--------|
| 5 We do it all the time, try to find new products and species to use. People must learn to eat more species | 5     | O      |

**Q 25 - credit**

How easy is it to gain access to credits, financing or loans for investments

[1–5; from 1 – very difficult, to 5 – very easy]

| Answer                                                                                                                 | Score Method |   |
|------------------------------------------------------------------------------------------------------------------------|--------------|---|
| 1 If you want to borrow 1 million for a machine, you need to have 1 million in your pocket. Not many fishers have that | 1            | O |

**Q 26 - unc\_23\_25**

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

| Answer                  | Score Method |   |
|-------------------------|--------------|---|
| 5 Very large variation. | 5            | O |

**Q 27 - knowledge\_transmission**

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

| Answer | Score Method |   |
|--------|--------------|---|
| 5      | 5            | O |

**Q 28 - benefit\_local\_community**

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                              | Score Method |   |
|-----------------------------------------------------------------------------------------------------|--------------|---|
| 5 Vendace fishing is extremely important for Kalix, not only for jobs but also for the town's image | 5            | O |

**Q 29 - benefit\_personal\_community**

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

| Answer | Score Method |   |
|--------|--------------|---|
| 5      | 5            | O |



**Q 30 - traditional\_food**

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                            | Score Method |   |
|---------------------------------------------------|--------------|---|
| [5] Very important: vendace, salmon, and herring. | 5            | O |

**Q 31 - unc\_26\_30**

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

| Answer                                                           | Score Method |   |
|------------------------------------------------------------------|--------------|---|
| [1] No variation. It is very important for everyone, I dare say. | 1            | O |

**Q 32 - outdoor\_benefit**

How important is it for you to be outdoors (when you fish) for your well-being

[1–5; from 1 – not important at all, to 5 – very important]

| Answer      | Score Method |   |
|-------------|--------------|---|
| 4 important | 4            | O |

**Q 33 - identity**

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

| Answer         | Score Method |   |
|----------------|--------------|---|
| [4] Important. | 4            | O |

**Q 34 - home**

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

| Answer                                                                                                                 | Score Method |   |
|------------------------------------------------------------------------------------------------------------------------|--------------|---|
| Hard to say for specific places. For example, Kalix archipelago what should I say? We live here, so yes, important.(4) | 4            | O |

**Q 35 - unc\_31\_34**

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

| Answer                                                                                                                                                       | Score | Method |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| [1] Not much variation. Most are very proud to be fishers; how much one fishes in practice may vary, but in terms of meaning, there are no major differences | 2     | G      |

**Q 36 - fdays**

Approximately how many days per year do you fish?

[free text description]

| Answer                                                                                                                                   | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit |
|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|------|
| Hard to say. If we have 6 months without ice, we actively fish 1 third of the days. But we keep the gear in the water all ice free days. | 50                        | 100                      | days |

**Q 37 - income**

Approximately how large is your gross income from a fishing day on average?

[free text description]

| Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|------------|
| Impossible to calculate per day. Over 6 months? It's the four weeks of vendace fishing that make up the income. This varies. I haven't broken down exactly what the fishing itself gives, plus there's all the processing afterwards. Then you'd also have to deduct labor costs. Teija calculates roughly to get a value: 1.4 million SEK per year from vendace fishing. Plus some perch and bream. We haven't fished salmon for several seasons — the quota is fished up in 7 days [by those who have good fishing waters near the Torne River], so it's not worth setting the gear, since it takes a week to set and a week to haul. | 1e+06                     | 1500000                  | sek_season |

**Q 38 - season\_maint**

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

| Answer | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit |
|--------|---------------------------|--------------------------|------|
|        |                           |                          |      |

|                                                                                                                                                                                                                               |       |       |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|------------|
| I don't remember, it gets lumped together with all other repairs at Guldhaven. We buy 1 boat per season several hundred thousand SEK and a new trawl, 200k SEK, other stuff 100k SEK. Roughly 0.5 million SEK over two years. | 1e+05 | 5e+05 | sek_season |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|------------|

### Q 39 - fuel\_trip

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

| Answer               | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit    |
|----------------------|---------------------------|--------------------------|---------|
| Don't dare estimate. | NA                        | NA                       | sek_day |

### Q 40 - extra\_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

| Answer                                       | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit    |
|----------------------------------------------|---------------------------|--------------------------|---------|
| Marginal, since we have such short distances | NA                        | NA                       | sek_day |

### Q 41 - extra\_crew

What are the costs for extra crew on board?

[free text description]

| Answer                                                                                                                                                                            | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|---------|
| None for fishing itself. We bring in staff for processing [130 people). Finding staff is the limiting factor in fishing, especially when there is as much vendace as there is now | NA                        | NA                       | sek_day |

### Q 42 - extra\_cleaning\_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

| Answer | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit |
|--------|---------------------------|--------------------------|------|
|        |                           |                          |      |

|                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|--------------|
| Cleaning traps takes some time, and it gets worse as the water gets warmer. Cleaning 10 traps took 0.5 h. Now we only have 20 traps in the water because of the perch, but when we fish the most, we have 100 traps. Then it means a lot of driving to clean them every three weeks; otherwise, they stay all season, or we clean once. Built ourselves from mink wire. Clean with a pressure washer. Same with bream traps | 3 | 3 | minutes_gear |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|--------------|

### Q 43 - exytra\_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?  
[free text description]

| Answer                                                                                                        | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit         |
|---------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|--------------|
| Things blow away. Doesn't cost much, but takes time to pick everything up when it's scattered across the bay. | NA                        | NA                       | not_reported |

### Q 44 - unc\_36\_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

| Answer | Score | Method |
|--------|-------|--------|
| 5      | 5     | O      |

### Q 45 - NA

Anything you want to add?

[free text description]

| Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Score    | Method |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| The striped perch are real stripes! They have found more in test fishing. Small ones, now frozen, which she will send to us. 234 perch in the test fishing, 9 33 cm. We have a lot of perch here now, never been so many on rod fishing — in a 1 hour round we get 10 perch of eating size. Why do you think? Now that the vendace stock is so good, they have a lot of food, and they grow well when it's warm. Seems to be good recruitment too, since we caught so many small ones. We had the nets here in Långviken, along a gradient from open sea to inner bay: mostly bream in the innermost net, and mostly perch in the outermost net. The largest perch were farthest out. | -<br>999 | NA     |

### Interviewee: I 3

3. name: dennis and lars-göran. category: coastal fisherman id3 is professional coastal fishermen in southwest bothnian sea (norrsundet). they fish with gillnets and traps. traps are only used for salmon. the main target is herring, whitefish, salmon, perch, and eel. they also run a restaurant and have several fish trucks. the herring fishing takes place a few hundred meters from the harbor. they mention

whitefish, salmon, perch in the local area, and in the gulf of Gävle. their average distance travel is about 1 hour (assuming 10 knots it is about 20 km) and the longest distance travel is 3 hours (assuming 10 knots it is max 55 km) the main fishing season is summer (from april to august). in winter they generally do not go out until spring comes. how they adapt: id3 can adapt when fishing with nets to hw by avoiding hottest hours, performing shorter sets and using cooling boxes. can also adapt to hw, ab and gales by changing fishing location. to avoid hw and ab they can go further out, the opposite for gales. in any case they can move on short distances and can cope with gales only in shallow waters, they don't go deep and with storms they don't go out at all. their spatial mobility is mainly affected by the boat (12m boat) and infrastructure (they want to go back home) on second instance by management that makes possible to fish in proximity (over a coastline spanning of about 60 km in aerial distance). there is not the same flexibility for salmon traps, which are fixed.

## Q 2 - go\_out

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1–5; from 1 – very unlikely, to 5 – very likely]

|   | Event                                     | Answer                                                                                                                                                                                                                                                 | Score | Method |
|---|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | Depends on ice. Boats are usually hauled up in winter. If forecast shows winter is over, we might launch. Unlikely to put in and haul up again. Might use the small boat if open water, but ice may return. Large boats remain on land. =1 regardless. | -999  | NA     |
| b | Heatwave in the summer season             | Then we fish, just go further out where it's cooler. =5.                                                                                                                                                                                               | 5     | O      |
| c | Severe icing (in the winter season)       | 5 (icing can happen in spring).                                                                                                                                                                                                                        | -999  | G      |
| d | Algal bloom (in the summer season)        | Rare up here, but still fish. =5.                                                                                                                                                                                                                      | 5     | O      |
| e | Extreme cloudburst (in the summer season) | If gear is already out, we go anyway. If not, we wait. =3.                                                                                                                                                                                             | 3     | O      |
| f | Gale                                      | No, we stay home. =1.                                                                                                                                                                                                                                  | 1     | O      |
| g | Storm                                     | No, we stay home. =1.                                                                                                                                                                                                                                  | 1     | O      |

## Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better]

|   | Event                         | Answer                       | Score | Method |
|---|-------------------------------|------------------------------|-------|--------|
| a | Heatwave in the winter season | Not relevant                 | -999  | NA     |
| b | Heatwave in the summer season | Heatwave makes it worse. =2. | 2     | O      |

|   |                                           |                                                                                                                                                      |      |    |
|---|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------|----|
| c | Severe icing (in the winter season)       | not relevant                                                                                                                                         | -999 | NA |
| d | Algal bloom (in the summer season)        | For perch, it gets worse (=2). Same for herring, due to warmth.                                                                                      | 3    | M  |
| e | Extreme cloudburst (in the summer season) | Doesn't affect catchability (only whether we go out). =3.                                                                                            | 3    | O  |
| f | Gale                                      | Yes – after a gale from the right direction, herring can blow in, salmon may swim more too (E, NE in summer). =4. From W & NW it gets worse instead. | NA   | M  |
| g | Storm                                     | Same, just stronger wind.                                                                                                                            | NA   | NA |

Multiple options identified

|   | Event                              | target  | Score (given) |
|---|------------------------------------|---------|---------------|
| d | Algal bloom (in the summer season) | perch   | 2             |
| d | Algal bloom (in the summer season) | herring | 2             |
| f | Gale                               | herring | 999           |
| f | Gale                               | salmon  | 999           |
| g | Storm                              | herring | 999           |
| g | Storm                              | salmon  | 999           |

#### Q 4 - catch\_assemblage

Do you catch a different composition of species during such extreme weather events?

[1–5; from 1 – very similar, to 5 – very different species composition]

|   | Event                                     | Answer           | Score | Method |
|---|-------------------------------------------|------------------|-------|--------|
| a | Heatwave in the winter season             | Not relevant.    | -999  | NA     |
| b | Heatwave in the summer season             | very similar (1) | 1     | O      |
| c | Severe icing (in the winter season)       | not relevant     | -999  | NA     |
| d | Algal bloom (in the summer season)        | not directly (1) | 1     | O      |
| e | Extreme cloudburst (in the summer season) | no (1)           | 1     | O      |
| f | Gale                                      | 1                | 1     | O      |

|   |       |   |   |   |
|---|-------|---|---|---|
| g | Storm | 1 | 1 | O |
|---|-------|---|---|---|

### Q 5 - catch\_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

|   | Event                                     | Answer                                                   | Score | Method |
|---|-------------------------------------------|----------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | not relevant                                             | -999  | NA     |
| b | Heatwave in the summer season             | much worse, must get them quickly into cooling boxes (1) | 1     | O      |
| c | Severe icing (in the winter season)       | not relevant                                             | -999  | NA     |
| d | Algal bloom (in the summer season)        | no, just from the heat                                   | 3     | NA     |
| e | Extreme cloudburst (in the summer season) | no (3)                                                   | 3     | O      |
| f | Gale                                      | no (3)                                                   | 3     | O      |
| g | Storm                                     | no (3)                                                   | 3     | O      |

### Q 6 - cope\_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1–5; from 1 – very unlikely, to 5 – very likely]

|   | Event                                     | Answer                                                                                                                 | Score | Method |
|---|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | NA                                                                                                                     | -999  | NA     |
| b | Heatwave in the summer season             | Only b. Go out before midday heat. Early mornings for herring, perch later in day. Adjust by hauling nets earlier. =5. | 5     | O      |
| c | Severe icing (in the winter season)       | NA                                                                                                                     | -999  | NA     |
| d | Algal bloom (in the summer season)        | NA                                                                                                                     | -999  | NA     |
| e | Extreme cloudburst (in the summer season) | NA                                                                                                                     | -999  | NA     |
| f | Gale                                      | NA                                                                                                                     | -999  | NA     |
| g | Storm                                     | NA                                                                                                                     | -999  | NA     |

**Q 7 - lag\_effect**

Do this extreme weather events have long-term effects on how you fish or catch composition

1

|   | Event                                     | Answer | Score | Method |
|---|-------------------------------------------|--------|-------|--------|
| a | Heatwave in the winter season             | NA     | -999  | NA     |
| b | Heatwave in the summer season             | NA     | 1     | NA     |
| c | Severe icing (in the winter season)       | NA     | -999  | NA     |
| d | Algal bloom (in the summer season)        | NA     | 1     | NA     |
| e | Extreme cloudburst (in the summer season) | NA     | 1     | NA     |
| f | Gale                                      | NA     | 1     | NA     |
| g | Storm                                     | NA     | 1     | NA     |

**Q 8 - unc\_2\_7**

How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? 1–5; from 1 – not at all, to 5 – very much

| Answer            | Score | Method |
|-------------------|-------|--------|
| 1 in Bothnian sea | 1     | O      |

**Q 9 - personal\_safety**

How much concern for your personal safety do these extreme weather events bring?

[0–2; 0 – no concern, 1 – personal injuries, 2 – life-threatening

|   | Event                               | Answer | Score | Method |
|---|-------------------------------------|--------|-------|--------|
| a | Heatwave in the winter season       | NA     | -999  | NA     |
| b | Heatwave in the summer season       | 0      | 0     | O      |
| c | Severe icing (in the winter season) | NA     | -999  | NA     |
| d | Algal bloom (in the summer season)  | 0      | 0     | O      |



|   |                                                 |                                          |      |    |
|---|-------------------------------------------------|------------------------------------------|------|----|
| e | Extreme<br>cloudburst (in the<br>summer season) | 0                                        | 0    | O  |
| f | Gale                                            | rarely go out but sudden events possible | 1    | NA |
| g | Storm                                           | don't go out                             | -998 | NA |

### Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0-3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

|   | Event                                           | Answer                                                   | Score | Method |
|---|-------------------------------------------------|----------------------------------------------------------|-------|--------|
| a | Heatwave in the<br>winter season                | NA                                                       | -999  | NA     |
| b | Heatwave in the<br>summer season                | NA                                                       | 0     | NA     |
| c | Severe icing (in<br>the winter season)          | NA                                                       | -999  | NA     |
| d | Algal bloom (in<br>the summer<br>season)        | 0, though nets can get dirty.                            | 0     | O      |
| e | Extreme<br>cloudburst (in the<br>summer season) | 0                                                        | 0     | O      |
| f | Gale                                            | Could damage or destroy gear/boat, or cause minor damage | 999   | NA     |
| g | Storm                                           | Could damage or destroy gear/boat, or cause minor damage | 999   | NA     |

### Q 11 - minor\_damage\_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

| Answer                                                                           | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit     |
|----------------------------------------------------------------------------------|---------------------------|--------------------------|----------|
| 50,000 SEK/year for nets that break, or propellers when<br>blown onto land/rocks | 1                         | 50000                    | sek_year |

### Q 12 - major\_damage\_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

|  |
|--|
|  |
|--|

| Answer                                                                                                              | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit |
|---------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|------|
| Rare for boats – then we wouldn't be out. But a salmon trap can be destroyed in a summer storm, costing 200,000 SEK | 50000                     | 2e+05                    | sek  |

### Q 13 - change\_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0–2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

|   | Event                                     | Answer                                                                                                   | Score | Method |
|---|-------------------------------------------|----------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | NA                                                                                                       | -999  | NA     |
| b | Heatwave in the summer season             | Yes, set in other places, change a lot. =2. Salmon gear is fixed at paid spots all summer, so they stay. | 2     | O      |
| c | Severe icing (in the winter season)       | NA                                                                                                       | -999  | NA     |
| d | Algal bloom (in the summer season)        | In bays, same as in heatwaves                                                                            | 2     | NA     |
| e | Extreme cloudburst (in the summer season) | 0                                                                                                        | 0     | O      |
| f | Gale                                      | 2. Move to less windy spots. For herring it's harder, so maybe stay home.                                | 2     | O      |
| g | Storm                                     | Don't go out at all, follow forecasts. Our boat is 12 m, so never in storms                              | -998  | NA     |

### Q 14 - unc\_9\_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

| Answer      | Score | Method |
|-------------|-------|--------|
| Uniform. 1. | 1     | O      |

### Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

|   | Event                         | Answer | Score | Method |
|---|-------------------------------|--------|-------|--------|
| a | Heatwave in the winter season | NA     | -999  | NA     |

|   |                                           |                                                                  |      |    |
|---|-------------------------------------------|------------------------------------------------------------------|------|----|
| b | Heatwave in the summer season             | Works – go further out (usually warmest inshore). =5.            | 5    | O  |
| c | Severe icing (in the winter season)       | NA                                                               | -999 | NA |
| d | Algal bloom (in the summer season)        | If heat lasts long, algal bloom spreads and worsens fishing. =2. | 2    | O  |
| e | Extreme cloudburst (in the summer season) | Yes, possible.                                                   | 4    | NA |
| f | Gale                                      | Yes, but usually means staying home                              | 4    | NA |
| g | Storm                                     | Storms are hard to avoid.                                        | 2    | NA |

### Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

| Answer                                                                                                                                                                                                                                              | Score | Method |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| Yes, can move around for herring, whitefish, perch. Only salmon gear is fixed. Our operations use active gear, so we can relocate, but only short distances – catch must be landed quickly. (Older generations even traveled to Trosa/Grisslehamn.) | 999   | NA     |

### Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer                                                  | Score | Method |
|---------------------------------------------------------|-------|--------|
| No – no infrastructure elsewhere, must return home. =0. | 0     | O      |

### Q 18 - management\_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer                                                                              | Score | Method |
|-------------------------------------------------------------------------------------|-------|--------|
| Not well informed. Probably possible in Gävle Bay (we have exemptions locally). =1. | 1     | O      |

**Q 19 - management\_species**

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer                                                       | Score Method |   |
|--------------------------------------------------------------|--------------|---|
| 0. Few commercial species here, license not species-specific | 0            | O |

**Q 20 - unc\_15\_19**

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

| Answer                     | Score Method |   |
|----------------------------|--------------|---|
| Same situation for all. =1 | 1            | O |

**Q 21 - stress**

How much stress does the risk of the following events cause (in general)?

[1–5; from 1 – really not stressful, to 5 – very stressful]

|   | Type             | Answer                                                                                                         | Score Method |   |
|---|------------------|----------------------------------------------------------------------------------------------------------------|--------------|---|
| a | stress_injuries  | 1, not stressed about that.                                                                                    | 1            | O |
| b | stress_gear      | Yes – after summer storms worried salmon traps may blow away. =5.                                              | 5            | O |
| c | stress_condition | Can happen if winds last days, then quality is lost, fish becomes worthless. Usually not stormy that long. =3. | 3            | O |

**Q 22 - unc\_21**

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

| Answer                                                     | Score Method |   |
|------------------------------------------------------------|--------------|---|
| Fishers are calm and experienced, know when to go out. =1. | 1            | O |

**Q 23 - job\_mobility**

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

| Answer | Score Method |  |
|--------|--------------|--|
|--------|--------------|--|

Yes – we already do many other things. Impossible to live solely off fishing on the Norrland coast. Must diversify, requires entrepreneurship. =5

## Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

| Answer                                                                                                                       | Score | Method |
|------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| Always striving to improve, never want to stop developing. Every time we smoke fish, want it better. Always want renewal. =5 | 5     | O      |

## Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1–5; from 1 – very difficult, to 5 – very easy]

| Answer                                                                                                                                 | Score | Method |
|----------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| 1. Not a thriving industry. We've thought about new boats for better work environment, but economically impossible with so little fish | 1     | O      |

## Q 26 - unc\_23\_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

| Answer                                                                                                                                          | Score | Method |
|-------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| Not all alike – some more ambitious than others, depends on vision. Some are “fair-weather fishers,” others try to make it work year-round. =5. | 5     | O      |

## Q 27 - knowledge\_transmission

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                                                                                                                                                                          | Score | Method |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| Very important. We want it to live on. Almost dying out, barely any succession in families. In Gävle municipality soon only us left. =5. That's why I started – to continue and develop what earlier generations built. History carries weight. | 5     | O      |

### Q 28 - benefit\_local\_community

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                                                                                                                                                                 | Score | Method |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| Very important, =5. We're the largest employer here. In July we had 50 on payroll. Coastal fishing benefits many. Attracts 40–50,000 tourists annually. They fuel up, shop at local stores. Huge significance for the whole community. | 5     | O      |

### Q 29 - benefit\_personal\_community

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                                                                                            | Score | Method |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| 5. Doesn't have to be me, but important that I'm the youngest investing – inspires cousins to invest too. Many long-time fishers easily deliver fish to Bergmans. | 5     | O      |

### Q 30 - traditional\_food

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                         | Score | Method |
|------------------------------------------------|-------|--------|
| 5. Very important for Swedish food production. | 5     | O      |

### Q 31 - unc\_26\_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

| Answer                                                                                                              | Score | Method |
|---------------------------------------------------------------------------------------------------------------------|-------|--------|
| Very similar. That's why they continue, try to recruit new ones. But hard since chances are poor, little money. =1. | 1     | O      |

### Q 32 - outdoor\_benefit

How important is it for you to be outdoors (when you fish) for your well-being

[1–5; from 1 – not important at all, to 5 – very important]

| Answer | Score | Method |
|--------|-------|--------|
| 5      | 5     | O      |

### Q 33 - identity

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

| Answer                                                                                           | Score Method |   |
|--------------------------------------------------------------------------------------------------|--------------|---|
| 5. Now it's my whole identity (used to play hockey, now entrepreneurship in fishing defines me). | 5            | O |

### Q 34 - home

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

| Answer                                                                              | Score Method |   |
|-------------------------------------------------------------------------------------|--------------|---|
| Very important – many specific places with memories. As important as self-image. =5 | 5            | O |

### Q 35 - unc\_31\_34

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

| Answer                                                                                                           | Score Method |    |
|------------------------------------------------------------------------------------------------------------------|--------------|----|
| Quite similar – everyone loves being outdoors, values the freedom. Calm water, sun shining – few jobs beat that! | 2            | NA |

### Q 36 - fdays

Approximately how many days per year do you fish?

[free text description]

| Answer                               | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit |
|--------------------------------------|---------------------------|--------------------------|------|
| Varies, 120–150 days of net fishing. | 120                       | 150                      | days |

### Q 37 - income

Approximately how large is your gross income from a fishing day on average?

[free text description]

| Answer                                                                     | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit    |
|----------------------------------------------------------------------------|---------------------------|--------------------------|---------|
| Hard to say. Some days 10–15,000 SEK, other days 0 if seals eat everything | 10000                     | 15000                    | sek_day |

**Q 38 - season\_maint**

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

| Answer                                                                                                                   | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit       |
|--------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|------------|
| About 200,000 SEK. 100–150,000 in damages. A lot of time spent on repairs, much voluntary work, patching holes in winter | 150000                    | 2e+05                    | sek_season |

**Q 39 - fuel\_trip**

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

| Answer                                                                                                                                     | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit    |
|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|---------|
| Several thousand SEK. Longest trips 2–3 hours out: 3,000–5,000 SEK/day. Short trips about half. Sometimes just outside the restaurant here | 1500                      | 2500                     | sek_day |

**Q 40 - extra\_fuel**

What are the costs for extra fuel to travel further than normal?

[free text description]

| Answer       | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit    |
|--------------|---------------------------|--------------------------|---------|
| See previous | 3000                      | 5000                     | sek_day |

**Q 41 - extra\_crew**

What are the costs for extra crew on board?

[free text description]

| Answer                                                                                             | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit     |
|----------------------------------------------------------------------------------------------------|---------------------------|--------------------------|----------|
| Rare – usually only if a lot of fish. Then about 200 SEK/hour, unless pensioners help voluntarily. | 0                         | 200                      | sek_hour |



**Q 42 - extra\_cleaning\_cost**

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

| Answer                                                                                                                                                                                                                    | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|-------------|
| 20–30 hours. Salmon traps: ~20 h each to clean at season end, sometimes more. Warm July brings more growth. In 2018 it was so warm that when we hauled a trap, with so much fouling, we nearly sank when a waterspout hit | 20                        | 30                       | hours_event |

**Q 43 - exytra\_other**

Are there any other extra costs that may arise during extreme weather events, and how large are they then?

[free text description]

| Answer                                       | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit         |
|----------------------------------------------|---------------------------|--------------------------|--------------|
| Mostly covered – repairs, gear blown ashore. | NA                        | NA                       | not_reported |

**Q 44 - unc\_36\_42**

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

| Answer                                               | Score | Method |
|------------------------------------------------------|-------|--------|
| No, not much variation. We use the same type of gear | 2     | NA     |

**Q 45 - NA**

Anything you want to add?

[free text description]

| Answer                                                                                                                      | Score    | Method |
|-----------------------------------------------------------------------------------------------------------------------------|----------|--------|
| No, I think everything's covered. Biggest problems are seals and industrial fishing. With so few fish, seals get aggressive | -<br>999 | NA     |

**Interviewee: I 4**

- name: kjell malmLöf. category: archipelago fisherman id4 is a resident of the archipelago in the northern proper baltic (hasselö, tjust archipelago). archipelago fisherman cannot sell their catchers. in the past, there was a very diversified fishing community with various types of nets for coastal and freshwater species. nowadays they continue to fish with nets for perch, herring, whitefish. he mentions also the importances of being able to catch fish with rods, but in the second interview he only refers to fishing with nets. fishing mostly happens during summer (april to august), winter fishing is less common but can happen. he mentions that if the water is free from ice they can cast nets also in winter. how they

adapt: id4 is not economically dependent on fishing and in general tend to do it only when there are good conditions. he can adapt to hw by performing shorter (not overnight) sets. can also adapt to hw, ab and storms by changing fishing location. to avoid hw and ab they can go further out, the opposite for storms. in any case they can move on very short distances. their spatial mobility is mainly affected by its assigned fishing lot.

## Q 2 - go\_out

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1–5; from 1 – very unlikely, to 5 – very likely]

|   | Event                                     | Answer                                                                                                                                                                                                 | Score | Method |
|---|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | Quite likely, if the water is open we'll go out, 4                                                                                                                                                     | 4     | O      |
| b | Heatwave in the summer season             | 2. Depends on how it affects winds and algal blooms; this year we had huge problems with algal blooms (1 month).                                                                                       | 2     | O      |
| c | Severe icing (in the winter season)       | No problem for us, too far south.                                                                                                                                                                      | -999  | NA     |
| d | Algal bloom (in the summer season)        | 1                                                                                                                                                                                                      | 1     | O      |
| e | Extreme cloudburst (in the summer season) | We can get to the boathouse, but from there out onto the lake it's more difficult; 1, if we don't have gear in the water. But we check the weather forecast very carefully before putting in the gear. | 1     | O      |
| f | Gale                                      | 1                                                                                                                                                                                                      | 1     | O      |
| g | Storm                                     | 1                                                                                                                                                                                                      | 1     | O      |

## Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better]

|   | Event                               | Answer                                                                                                                                     | Score | Method |
|---|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season       | NA                                                                                                                                         | 2     | NA     |
| b | Heatwave in the summer season       | Yes, in the past when perch existed, they liked warmth, but now there are no perch, so the effect is not so big on the species we fish for | 3     | NA     |
| c | Severe icing (in the winter season) | NA                                                                                                                                         | -999  | NA     |

|   |                                           |                                                                                                                                                                                                                                                                                                                                                                                                   |   |    |
|---|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|
| d | Algal bloom (in the summer season)        | It's devastating, depressing to live around this algal soup, grandchildren can't swim, it's an imbalance in the Baltic Sea ecosystem, a sad new phenomenon. We try to deal with it when it drifts ashore. It becomes a white coating on the bottom when it sinks. We're worried about algal toxins (cows can break them down somewhat, but single-stomached animals can't; affects the liver). 1. | 1 | O  |
| e | Extreme cloudburst (in the summer season) | No, 3                                                                                                                                                                                                                                                                                                                                                                                             | 3 | O  |
| f | Gale                                      | A little worried about strong winds since we don't have big boats.                                                                                                                                                                                                                                                                                                                                | 2 | NA |
| g | Storm                                     | Same with storms.                                                                                                                                                                                                                                                                                                                                                                                 | 2 | NA |

#### Q 4 - catch\_assemblage

Do you catch a different composition of species during such extreme weather events?

[1-5; from 1 – very similar, to 5 – very different species composition]

|   | Event                                     | Answer                                                                            | Score | Method |
|---|-------------------------------------------|-----------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | Can't answer.                                                                     | -998  | NA     |
| b | Heatwave in the summer season             | Can't answer.                                                                     | -998  | NA     |
| c | Severe icing (in the winter season)       | NA                                                                                | -999  | NA     |
| d | Algal bloom (in the summer season)        | We don't really fish much then; would guess the fish flee from the algal bloom, 5 | 5     | O      |
| e | Extreme cloudburst (in the summer season) | Can't answer                                                                      | -998  | NA     |
| f | Gale                                      | Hard to say, we don't fish much then                                              | -998  | NA     |
| g | Storm                                     | Hard to say, we don't fish much then                                              | -998  | NA     |

#### Q 5 - catch\_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1-5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

|   | Event                         | Answer                                                                           | Score | Method |
|---|-------------------------------|----------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season | Don't think it affects condition, 3.                                             | 3     | O      |
| b | Heatwave in the summer season | hen the water is warmer, fish flesh becomes looser (especially fattier fish), 2. | 2     | O      |

|   |                                           |                    |      |    |
|---|-------------------------------------------|--------------------|------|----|
| c | Severe icing (in the winter season)       | we don't fish then | -999 | NA |
| d | Algal bloom (in the summer season)        | we don't fish then | -998 | NA |
| e | Extreme cloudburst (in the summer season) | we don't fish then | -998 | NA |
| f | Gale                                      | we don't fish then | -998 | NA |
| g | Storm                                     | we don't fish then | -998 | NA |

### Q 6 - cope\_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1–5; from 1 – very unlikely, to 5 – very likely]

|   | Event                                     | Answer                                                                                                                                                                                                                         | Score | Method |
|---|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | NA                                                                                                                                                                                                                             | -999  | NA     |
| b | Heatwave in the summer season             | If perch turn yellow and bad from the heat, we can do a day-set (not overnight). Have also set nets around a shoal and driven fish inside, which improved quality. We don't have ice machines, so we can influence it a little | 3     | NA     |
| c | Severe icing (in the winter season)       | NA                                                                                                                                                                                                                             | -999  | NA     |
| d | Algal bloom (in the summer season)        | NA                                                                                                                                                                                                                             | -999  | NA     |
| e | Extreme cloudburst (in the summer season) | NA                                                                                                                                                                                                                             | -999  | NA     |
| f | Gale                                      | NA                                                                                                                                                                                                                             | -999  | NA     |
| g | Storm                                     | NA                                                                                                                                                                                                                             | -999  | NA     |

### Q 7 - lag\_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

Algal bloom doesn't need more comment, it's a big problem we've already discussed. Other factor is wind direction, NW is bad, then the fish disappear. Big impact. This spring it locked in at N & NW with near gale winds for several weeks

|   | Event                         | Answer | Score | Method |
|---|-------------------------------|--------|-------|--------|
| a | Heatwave in the winter season | NA     | -999  | NA     |

|   |                                           |    |      |    |
|---|-------------------------------------------|----|------|----|
| b | Heatwave in the summer season             | NA | -999 | NA |
| c | Severe icing (in the winter season)       | NA | -999 | NA |
| d | Algal bloom (in the summer season)        | NA | 5    | NA |
| e | Extreme cloudburst (in the summer season) | NA | -999 | NA |
| f | Gale                                      | NA | 5    | NA |
| g | Storm                                     | NA | 5    | NA |

#### Q 8 - unc\_2\_7

How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? 1–5; from 1 – not at all, to 5 – very much

| Answer                                                                                                  | Score | Method |
|---------------------------------------------------------------------------------------------------------|-------|--------|
| Among subsistence fishers there's quite strong agreement; we communicate fairly well with each other, 2 | 2     | O      |

#### Q 9 - personal\_safety

How much concern for your personal safety do these extreme weather events bring?

[0–2; 0 – no concern, 1 – personal injuries, 2 – life-threatening

With strong winds, you adapt as an islander. I also do forestry, and if weather is bad with fog and strong winds, you plan activities so as not to risk injury. You adapt to weather

|   | Event                                     | Answer                                    | Score | Method |
|---|-------------------------------------------|-------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | no worry                                  | 0     | NA     |
| b | Heatwave in the summer season             | no worry                                  | 0     | NA     |
| c | Severe icing (in the winter season)       | NA                                        | -999  | NA     |
| d | Algal bloom (in the summer season)        | avoid eating fish caught during bloom     | 0     | NA     |
| e | Extreme cloudburst (in the summer season) | no worry, just wear the right clothes     | 0     | NA     |
| f | Gale                                      | slipping, breaking a leg from the rocking | 1     | NA     |
| g | Storm                                     | slipping, breaking a leg from the rocking | 1     | NA     |

### Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

|   | Event                                     | Answer                                                                                               | Score | Method |
|---|-------------------------------------------|------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | 0                                                                                                    | 0     | O      |
| b | Heatwave in the summer season             | with low water (running aground), 1                                                                  | 1     | O      |
| c | Severe icing (in the winter season)       | NA                                                                                                   | -999  | NA     |
| d | Algal bloom (in the summer season)        | 0                                                                                                    | 0     | O      |
| e | Extreme cloudburst (in the summer season) | 0                                                                                                    | 0     | O      |
| f | Gale                                      | Gear can drag and be damaged, very rare; boat damage when docking can be critical, minor repairs, 1. | 1     | O      |
| g | Storm                                     | Gear can drag and be damaged, very rare; boat damage when docking can be critical, minor repairs, 1. | 1     | O      |

### Q 11 - minor\_damage\_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

| Answer                                                                                                    | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit     |
|-----------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|----------|
| Propeller replacement is quite heavy, costs 2,500 SEK. Hull repair we fix ourselves, a couple hundred SEK | 200                       | 2500                     | sek_unit |

### Q 12 - major\_damage\_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

| Answer                                                                                               | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit     |
|------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|----------|
| Engines and boats are insured; we must cover gear ourselves. A few thousand SEK more than the 2,500. | 2500                      | 5000                     | sek_unit |

### Q 13 - change\_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0–2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

|   | Event                                     | Answer                                                                                                                          | Score | Method |
|---|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | 0                                                                                                                               | 0     | O      |
| b | Heatwave in the summer season             | 0 Only with algal bloom. If we want whitefish in summer during heatwave, they move deeper (15 m), =1. Move a few hundred meters | 1     | O      |
| c | Severe icing (in the winter season)       | NA                                                                                                                              | -999  | NA     |
| d | Algal bloom (in the summer season)        | 2 then we do not go fish                                                                                                        | 2     | O      |
| e | Extreme cloudburst (in the summer season) | Don't go out to set nets, but we do collect them if already out                                                                 | 1     | NA     |
| f | Gale                                      | We sometimes set in more sheltered bays less exposed, =2.                                                                       | 2     | O      |
| g | Storm                                     | We sometimes set in more sheltered bays less exposed, =2.                                                                       | 2     | O      |

### Q 14 - unc\_9\_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

| Answer | Score | Method |
|--------|-------|--------|
| 1, no  | 1     | O      |

### Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

|   | Event                               | Answer                                                                                | Score | Method |
|---|-------------------------------------|---------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season       | no need                                                                               | -999  | NA     |
| b | Heatwave in the summer season       | Drive grandchildren to another bay; no, we don't change fishing spot, we stop fishing | 1     | NA     |
| c | Severe icing (in the winter season) | NA                                                                                    | -999  | NA     |
| d | Algal bloom (in the summer season)  | Stop fishing                                                                          | 1     | NA     |

|   |                                           |                           |   |    |
|---|-------------------------------------------|---------------------------|---|----|
| e | Extreme cloudburst (in the summer season) | 1, no, we stop completely | 1 | O  |
| f | Gale                                      | 4, switch bays            | 4 | O  |
| g | Storm                                     | Don't go fishing          | 1 | NA |

#### Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0-2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

| Answer      | Score | Method |
|-------------|-------|--------|
| 1, somewhat | 1     | O      |

#### Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0-2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer                                                                                               | Score | Method |
|------------------------------------------------------------------------------------------------------|-------|--------|
| Yes, but we don't, too much hassle. Since we're not economically dependent, we can skip fishing. =2. | 2     | O      |

#### Q 18 - management\_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0-2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer                                                                                                                            | Score | Method |
|-----------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| We have common fishing waters (a cooperative). Fishing lots have become permanent, so I stick to my lot. =0 (within cooperative). | 0     | O      |

#### Q 19 - management\_species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0-2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer | Score | Method |
|--------|-------|--------|
|--------|-------|--------|



|                                                                                                                                                                                                                         |     |    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|
| Eel fishing ban; voluntary fishing stop Mar 15–May 15 in some spawning bays. We must restore predatory fish stocks. We think fish shelters (we've built 10 over 10 years) have limited effect at such low stock levels. | 999 | NA |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|

### Q 20 - unc\_15\_19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

| Answer                                                                   | Score | Method |
|--------------------------------------------------------------------------|-------|--------|
| Requires a vehicle/tractor, not all have it, but roughly consistent, =2. | 2     | O      |

### Q 21 - stress

How much stress does the risk of the following events cause (in general)?

[1–5; from 1 – really not stressful, to 5 – very stressful]

| Type               | Answer                                                                        | Score | Method |
|--------------------|-------------------------------------------------------------------------------|-------|--------|
| a stress_injuries  | not much, but in winter it's cold, windy, if motors fail then big problem, 3. | 3     | O      |
| b stress_gear      | 1                                                                             | 1     | O      |
| c stress_condition | 1                                                                             | 1     | O      |

### Q 22 - unc\_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

| Answer                          | Score | Method |
|---------------------------------|-------|--------|
| Probably fairly consistent, =2. | 2     | O      |

### Q 23 - job\_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

| Answer    | Score | Method |
|-----------|-------|--------|
| Not Asked | -     | NA     |
|           | 999   |        |

### Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

| Answer                                                                                                 | Score Method |   |
|--------------------------------------------------------------------------------------------------------|--------------|---|
| Thinking about whether nets are best; with traps the fish are less affected. Just thoughts so far. =2. | 2            | O |

### Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1–5; from 1 – very difficult, to 5 – very easy]

| Answer    | Score Method |    |
|-----------|--------------|----|
| Not Asked | -            | NA |
|           | 999          |    |

### Q 26 - unc\_23\_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

| Answer                                                                                 | Score Method |    |
|----------------------------------------------------------------------------------------|--------------|----|
| Most are conservative, professionals are more innovative. In my group, low innovation. | 2            | NA |

### Q 27 - knowledge\_transmission

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                                                                    | Score Method |   |
|-------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|
| 5. We've lost local competence when people die and take knowledge with them (e.g., neighbor who knew when/where to catch turbot, burbot). | 5            | O |

### Q 28 - benefit\_local\_community

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                                         | Score Method |   |
|----------------------------------------------------------------------------------------------------------------|--------------|---|
| I can't sell/deliver fish, only the professional fisher can. More cultural—nice to have fish on the table. =1. | 1            | O |

### Q 29 - benefit\_personal\_community

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Score | Method  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------|
| Hard to judge. There have been big changes recently – many have passed away or disappeared, and some younger people have moved in who have the opportunity to fish but haven’t started yet, they’re still renovating. At our fishing meetings people do show up – they’re not uninterested. People are interested in fishing, but to what extent they actually fish or communicate about it varies. It creates a unifying factor among us subsistence fishers who belong to the village association. Those who own a joint property with fishing rights and waters – and then there are the summer guests, but I have no idea about them. | 3.    | O<br> 3 |

### Q 30 - traditional\_food

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                                                                                                                                                                                                                         | Score | Method |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| Generational gap. We like fried herring, but grandchildren want hamburgers. Son hates flounder (too much growing up). Daughter better. Have to teach them every time how to debone herring. 5. Also for preparedness—resilience groups forming on islands (root cellars, handling power cuts). | 5     | G      |

### Q 31 - unc\_26\_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

| Answer                                          | Score | Method |
|-------------------------------------------------|-------|--------|
| Most people think roughly as we have just said. | 2     | NA     |

### Q 32 - outdoor\_benefit

How important is it for you to be outdoors (when you fish) for your well-being

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                 | Score | Method |
|----------------------------------------------------------------------------------------|-------|--------|
| 5. During test fishing we had wonderful mornings lifting nets—that’s what you live on. | 5     | O      |

**Q 33 - identity**

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

| Answer                                                                       | Score Method |   |
|------------------------------------------------------------------------------|--------------|---|
| Not so much the fishing, but managing it (chair of fishing association). =3. | 3            | O |

**Q 34 - home**

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

| Answer                                                          | Score Method |   |
|-----------------------------------------------------------------|--------------|---|
| There are several places we often go. Has some meaning, about 3 | 3            | O |

**Q 35 - unc\_31\_34**

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

| Answer                      | Score Method |    |
|-----------------------------|--------------|----|
| Can't say, emotional things | -<br>998     | NA |

**Q 36 - fdays**

Approximately how many days per year do you fish?

[free text description]

| Answer                          | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit |
|---------------------------------|---------------------------|--------------------------|------|
| Not much lately, maybe 10 days. | 10                        | NA                       | days |

**Q 37 - income**

Approximately how large is your gross income from a fishing day on average?

[free text description]

| Answer | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit         |
|--------|---------------------------|--------------------------|--------------|
| NA     | NA                        | NA                       | not_reported |

**Q 38 - season\_maint**

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

| Answer | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit       |
|--------|---------------------------|--------------------------|------------|
| 5k SEK | NA                        | 5000                     | sek_season |

**Q 39 - fuel\_trip**

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

| Answer                                                                                                     | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit       |
|------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|------------|
| Significant; 20,000 SEK yearly, much for boats, transport to land, island transport. About 5,000 SEK/year. | 5000                      | 20000                    | sek_season |

**Q 40 - extra\_fuel**

What are the costs for extra fuel to travel further than normal?

[free text description]

| Answer | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit         |
|--------|---------------------------|--------------------------|--------------|
| NA     | NA                        | NA                       | not_reported |

**Q 41 - extra\_crew**

What are the costs for extra crew on board?

[free text description]

| Answer | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit         |
|--------|---------------------------|--------------------------|--------------|
| NA     | NA                        | NA                       | not_reported |

**Q 42 - extra\_cleaning\_cost**

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|--|--|--|--|

| Answer                                   | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit         |
|------------------------------------------|---------------------------|--------------------------|--------------|
| When cleaning fish, takes a lot of time. | NA                        | NA                       | not_reported |

#### Q 43 - exytra\_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?  
[free text description]

| Answer                                                                                                                                                                                                                             | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|--------------|
| I think we've gone through most of it. Water levels: your report mentions it, and many are concerned about low water levels, which indeed happened; but high water is just as problematic – wading out to the boathouse in waders. | NA                        | NA                       | not_reported |

#### Q 44 - unc\_36\_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

| Answer                                                                                                                                                                                                                                                       | Score | Method |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| There are some who have fishing rights but don't fish at all, and then some who fish like I do, just now and then. Then there are those who have leased out their fishing lot, and in those cases there is a subsistence fisher who fishes quite intensively | 5     | NA     |

#### Q 45 - NA

Anything you want to add?  
[free text description]

| Answer                                                                                                                                                                                                                                                                                                                           | Score    | Method |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| Regarding innovation: we're interested in all cooperation with experts who can help us restore ecological balance in the Baltic Sea. There is interest moving forward. What I pick up is that we must do something about the algal blooms. Maybe find a way to collect them with boats; there is centrifuge competence in Sweden | -<br>999 | NA     |

#### Interviewee: I 5

5. name: lars ljunggren. category: recreational fisherman id5 represents the organization sportfiskarna, which is active in all sweden. he is based in stockholm. the members of sportfiskarna are anglers that mostly fish for pike, perch and zander. additionally, fishing for sea trout and trolling for salmon is common. they fish in lakes, streams and coastal areas. they can use boats in coastal areas and lakes, or just fish from the shore. in winter they also do ice fishing on lakes and generally they do not use boats until the ice is completely gone. the members practice both catch-and-release and keeping the catch. recreational fisherman can move around different fishing spots. how they adapt: sportfisherman

are diverse but have in common to fish exclusively for fun. they tend to fish only when it is a pleasant experience. they can cope to some extent with extreme events by adapting their location if they want to fish on the coast (going deeper or finding sheltered places) or move to lakes. their spatial mobility has virtually no limits.

## Q 2 - go\_out

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1–5; from 1 – very unlikely, to 5 – very likely]

|   | Event                                                                                        | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Score | Method |
|---|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season                                                                | If it's otherwise a good winter, then it's just nice weather. But if it's a mild winter, then you can't go out. Is ice fishing the dominant form? Few go out in boats? On the big lakes. and at Simrishamn there is some trolling from boats. If we think about ice fishing? Absolutely, =1 if the ice isn't there or is dangerous to walk on.                                                                                                                                                     | 1     | O      |
| b | Heatwave in the summer season                                                                | Yes, but in another way. When it's hot, fishing in running waters (salmon, trout, grayling) is closed. Perch and zander fishing only gets better in warm weather. Few target pike in a heatwave. But if you booked a salmon-fishing week, it simply gets canceled. Zander & perch = 5; salmon & trout = 1; pike = 2 (or 1, discussed with another angler present). But "vacation fishers" will probably go out regardless. Pike fishing is usually in spring & autumn; high temps affect survival. | NA    | M      |
| c | Severe icing (in the winter season)                                                          | Big impact.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1     | NA     |
| d | Algal bloom (in the summer season)                                                           | Less attractive, but still possible to fish; =2.                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2     | O      |
| e | Extreme cloudburst (in the summer season)                                                    | People hold back; it's a comfort factor. Most stay indoors                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2     | NA     |
| f | Gale                                                                                         | Yes, people hold back; most stay home, except the really keen ones who may go out in sheltered spots                                                                                                                                                                                                                                                                                                                                                                                               | 2     | NA     |
| g | Storm                                                                                        | No, impossible to fish.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1     | NA     |
| h | Are there any of these in combination that affect how likely it is that you go out and fish? | Combination of gale/storm and heavy rain – Even more unpleasant, a negative synergy effect                                                                                                                                                                                                                                                                                                                                                                                                         | NA    | NA     |

Multiple options identified

|   | Event                         | target | Score (given) |
|---|-------------------------------|--------|---------------|
| b | Heatwave in the summer season | zander | 5             |

|   |                               |        |   |
|---|-------------------------------|--------|---|
| b | Heatwave in the summer season | perch  | 5 |
| b | Heatwave in the summer season | trout  | 1 |
| b | Heatwave in the summer season | salmon | 1 |
| b | Heatwave in the summer season | pike   | 2 |

### Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better]

|   | Event                                     | Answer                                                                                                                | Score | Method |
|---|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | If there's no ice, you can't catch them with an ice rod, and you don't put in a boat just for a short ice-free spell. | -997  | NA     |
| b | Heatwave in the summer season             | species that dislike heat (salmonids) and even pike become harder to catch; perch and zander improve                  | NA    | M      |
| c | Severe icing (in the winter season)       | Heavy icing is rare, and few troll in sub-zero temps – marginal effect on fish                                        | 2     | NA     |
| d | Algal bloom (in the summer season)        | Don't know                                                                                                            | -999  | NA     |
| e | Extreme cloudburst (in the summer season) | No.                                                                                                                   | 3     | NA     |
| f | Gale                                      | Makes fishing more difficult, but doesn't affect catchability                                                         | 2     | NA     |
| g | Storm                                     | Nobody goes out in storms, so we don't know if fish bite then                                                         | -999  | NA     |

Multiple options identified

|   | Event                         | target | Score (given) |
|---|-------------------------------|--------|---------------|
| b | Heatwave in the summer season | zander | 4             |
| b | Heatwave in the summer season | perch  | 4             |
| b | Heatwave in the summer season | trout  | 2             |
| b | Heatwave in the summer season | salmon | 2             |



#### Q 4 - catch\_assemblage

Do you catch a different composition of species during such extreme weather events?

[1–5; from 1 – very similar, to 5 – very different species composition]

|   | Event                                     | Answer                                                                                                                                                                                                                                            | Score | Method |
|---|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | No, very similar                                                                                                                                                                                                                                  | 1     | NA     |
| b | Heatwave in the summer season             | No, you fish where you fish, so you get the species present there. But in mountain areas, if you target trout or grayling but the system also has pike and perch, you might get more of those instead. From a boat it's usually small differences | 2     | NA     |
| c | Severe icing (in the winter season)       | No, icing is rare                                                                                                                                                                                                                                 | -998  | NA     |
| d | Algal bloom (in the summer season)        | No                                                                                                                                                                                                                                                | 1     | NA     |
| e | Extreme cloudburst (in the summer season) | No                                                                                                                                                                                                                                                | 1     | NA     |
| f | Gale                                      | Not directly, but on the coast with prolonged winds you can get upwelling; perch may move closer inshore, and trout & whitefish may appear shallower than normal. Seen within a season                                                            | 999   | NA     |
| g | Storm                                     | Not directly, but on the coast with prolonged winds you can get upwelling; perch may move closer inshore, and trout & whitefish may appear shallower than normal. Seen within a season                                                            | 999   | NA     |

#### Q 5 - catch\_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

|   | Event                               | Answer                                                         | Score | Method |
|---|-------------------------------------|----------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season       | no                                                             | 3     | NA     |
| b | Heatwave in the summer season       | Yes, species that thrive in cooler waters get poorer condition | 2     | NA     |
| c | Severe icing (in the winter season) | NA                                                             | -998  | NA     |
| d | Algal bloom (in the summer season)  | No (guess)                                                     | 3     | NA     |

|   |                                                 |    |   |    |
|---|-------------------------------------------------|----|---|----|
| e | Extreme<br>cloudburst (in the<br>summer season) | no | 3 | NA |
| f | Gale                                            | no | 3 | NA |
| g | Storm                                           | no | 3 | NA |

### Q 6 - cope\_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1–5; from 1 – very unlikely, to 5 – very likely]

|   | Event                                           | Answer                                                                                                                                                      | Score | Method |
|---|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the<br>winter season                | NA                                                                                                                                                          | -999  | NA     |
| b | Heatwave in the<br>summer season                | No, nothing can be done. If fish are in poor condition, there's nothing you can do. Doesn't really apply to sport fishing since fish aren't trapped in gear | 1     | NA     |
| c | Severe icing (in<br>the winter season)          | NA                                                                                                                                                          | -999  | NA     |
| d | Algal bloom (in<br>the summer<br>season)        | NA                                                                                                                                                          | -999  | NA     |
| e | Extreme<br>cloudburst (in the<br>summer season) | NA                                                                                                                                                          | -999  | NA     |
| f | Gale                                            | NA                                                                                                                                                          | -999  | NA     |
| g | Storm                                           | NA                                                                                                                                                          | -999  | NA     |

### Q 7 - lag\_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

Greatest effect is probably on Arctic char (though not a major sportfishing species). Other species may move in: grayling, trout, whitefish, perch & pike. Biggest sportfishing impact from extreme weather is likely on char. On the coast, fish can always go deeper. Does it affect how you fish? Yes – e.g., trout anglers (spring & autumn from shore) may need to fish deeper or at different times. On Gotland, sea trout fishing is big – if warming lasts longer, the season gets shorter until it cools again.

|   | Event                            | Answer | Score | Method |
|---|----------------------------------|--------|-------|--------|
| a | Heatwave in the<br>winter season | NA     | -999  | NA     |
| b | Heatwave in the<br>summer season | NA     | 3     | NA     |

|   |                                           |    |      |    |
|---|-------------------------------------------|----|------|----|
| c | Severe icing (in the winter season)       | NA | -999 | NA |
| d | Algal bloom (in the summer season)        | NA | -999 | NA |
| e | Extreme cloudburst (in the summer season) | NA | -999 | NA |
| f | Gale                                      | NA | -999 | NA |
| g | Storm                                     | NA | -999 | NA |

#### Q 8 - unc\_2\_7

How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? 1–5; from 1 – not at all, to 5 – very much

| Answer                                          | Score | Method |
|-------------------------------------------------|-------|--------|
| Depends on when/how you fish; varies a lot, =5. | 5     | O      |

#### Q 9 - personal\_safety

How much concern for your personal safety do these extreme weather events bring?

[0–2; 0 – no concern, 1 – personal injuries, 2 – life-threatening

|   | Event                                     | Answer                                                                                       | Score | Method |
|---|-------------------------------------------|----------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | for those fishing then, big concern – must assess ice conditions if a winter heatwave occurs | 2     | NA     |
| b | Heatwave in the summer season             | 0                                                                                            | 0     | O      |
| c | Severe icing (in the winter season)       | If someone is foolish enough to go out, some risk, but few do                                | 1     | NA     |
| d | Algal bloom (in the summer season)        | 0                                                                                            | 0     | O      |
| e | Extreme cloudburst (in the summer season) | 0                                                                                            | 0     | O      |
| f | Gale                                      | 0                                                                                            | 0     | O      |
| g | Storm                                     | Surprise storms can be worrisome 2                                                           | 2     | O      |

#### Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

|   | Event                                     | Answer                             | Score | Method |
|---|-------------------------------------------|------------------------------------|-------|--------|
| a | Heatwave in the winter season             | 0                                  | 0     | O      |
| b | Heatwave in the summer season             | 0                                  | 0     | O      |
| c | Severe icing (in the winter season)       | 1 (but most boats are stored then) | 1     | O      |
| d | Algal bloom (in the summer season)        | 0                                  | 0     | O      |
| e | Extreme cloudburst (in the summer season) | could sink a boat if unlucky, 3    | 3     | O      |
| f | Gale                                      | no                                 | 0     | NA     |
| g | Storm                                     | yes could sink a boat (3)          | 3     | O      |

#### Q 11 - minor\_damage\_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

| Answer                                                                            | Score - low (given) | Score - up (given) | Unit     |
|-----------------------------------------------------------------------------------|---------------------|--------------------|----------|
| Anything from 10,000 SEK to 1 million SEK – these are expensive toys; hard to say | 10000               | 50000              | sek_unit |

#### Q 12 - major\_damage\_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

| Answer                                                                                                                               | Score - low (given) | Score - up (given) | Unit     |
|--------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------|----------|
| Up to half a million SEK, not uncommon. For new trolling setups, even more. Sportfishing boats cost 50–300,000 SEK, sometimes double | 50000               | 1000000            | sek_unit |

#### Q 13 - change\_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0–2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

| Event | Answer | Score | Method |
|-------|--------|-------|--------|
|-------|--------|-------|--------|

|   |                                           |                                                                     |      |    |
|---|-------------------------------------------|---------------------------------------------------------------------|------|----|
| a | Heatwave in the winter season             | 2 – if the ice is unsafe, you don't go out.                         | 2    | O  |
| b | Heatwave in the summer season             | Mixed                                                               | NA   | NA |
| c | Severe icing (in the winter season)       | NA                                                                  | -998 | NA |
| d | Algal bloom (in the summer season)        | Maybe 1 – switch to another bay; if bloom is everywhere, then skip. | 1    | O  |
| e | Extreme cloudburst (in the summer season) | 2 – you stay indoors                                                | 2    | O  |
| f | Gale                                      | If sheltered, you can fish a little, =1, unless fully blown in.     | 1    | O  |
| g | Storm                                     | Same                                                                | 1    | O  |

Multiple options identified

|   | Event                         | target       | Score (given) |
|---|-------------------------------|--------------|---------------|
| b | Heatwave in the summer season | perch        | 0             |
| b | Heatwave in the summer season | salmon/trout | 2             |

#### Q 14 - unc\_9\_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

| Answer                                         | Score | Method |
|------------------------------------------------|-------|--------|
| Depends on type of fishing – varies a lot, =5. | 5     | O      |

#### Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

|   | Event                               | Answer                | Score | Method |
|---|-------------------------------------|-----------------------|-------|--------|
| a | Heatwave in the winter season       | Hard, scale is large. | 2     | NA     |
| b | Heatwave in the summer season       | Hard                  | 2     | NA     |
| c | Severe icing (in the winter season) | NA                    | -998  | NA     |

|   |                                           |                                                             |   |    |
|---|-------------------------------------------|-------------------------------------------------------------|---|----|
| d | Algal bloom (in the summer season)        | Easier – could switch to a lake if planning coastal fishing | 4 | NA |
| e | Extreme cloudburst (in the summer season) | Hard                                                        | 2 | NA |
| f | Gale                                      | to some extent – find a sheltered spot                      | 3 | NA |
| g | Storm                                     | no                                                          | 1 | NA |

### Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

| Answer                                                               | Score Method |   |
|----------------------------------------------------------------------|--------------|---|
| Most can be somewhat mobile. Some are keen enough to travel far. =1. | 1            | O |

### Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer                                                                                                 | Score Method |    |
|--------------------------------------------------------------------------------------------------------|--------------|----|
| For mobile sport fishers, infrastructure is crucial. It exists, but not always accessible for everyone | 1            | NA |

### Q 18 - management\_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer                                                                                                             | Score Method |   |
|--------------------------------------------------------------------------------------------------------------------|--------------|---|
| 2. But you may need to buy a fishing license for new spots. On the coast it's free, which allows a lot of freedom. | 2            | O |

### Q 19 - management\_species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer                                                                                                                                              | Score | Method |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| Some restrictions. On the coast, fairly free. But salmon, trout & grayling are closed during spawning, or when it gets too warm (quite common). =1. | 1     | O      |

#### Q 20 - unc\_15\_19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

| Answer                                                    | Score | Method |
|-----------------------------------------------------------|-------|--------|
| Quite a lot – some don't fish from boats at all, some do. | 4     | NA     |

#### Q 21 - stress

How much stress does the risk of the following events cause (in general)?

[1–5; from 1 – really not stressful, to 5 – very stressful]

| Type               | Answer                                                              | Score | Method |
|--------------------|---------------------------------------------------------------------|-------|--------|
| a stress_injuries  | Not really – only extreme cases (climbing steep riverbanks, pushing | 1     | NA     |
| b stress_gear      | no                                                                  | 1     | NA     |
| c stress_condition | no                                                                  | 1     | NA     |

#### Q 22 - unc\_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

| Answer                                        | Score | Method |
|-----------------------------------------------|-------|--------|
| some variation, but less than other questions | 2     | NA     |

#### Q 23 - job\_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

| Answer    | Score | Method |
|-----------|-------|--------|
| not asked | -     | NA     |
|           | 999   |        |

#### Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

| Answer                                       | Score Method |   |
|----------------------------------------------|--------------|---|
| I'm probably quite creative, I think so. =4. | 4            | O |

#### Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1–5; from 1 – very difficult, to 5 – very easy]

| Answer    | Score Method |    |
|-----------|--------------|----|
| not asked | -            | NA |
|           | 999          |    |

#### Q 26 - unc\_23\_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

| Answer                         | Score Method |    |
|--------------------------------|--------------|----|
| Varies greatly (only for Q24). | 4            | NA |

#### Q 27 - knowledge\_transmission

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                  | Score Method |   |
|-------------------------------------------------------------------------|--------------|---|
| Relatively important – nice if the next generation also enjoys fishing. | 4            | O |
| Fairly important. =4.                                                   |              |   |

#### Q 28 - benefit\_local\_community

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                                                                                                   | Score Method |    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----|
| Generally, not a strong link. But in rural areas, sport fishing can keep the village shop open due to visiting guests in summer. In rural areas it has real significance | 5            | NA |



**Q 29 - benefit\_personal\_community**

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                                                        | Score Method |    |
|-------------------------------------------------------------------------------------------------------------------------------|--------------|----|
| Some drive among those eager to show off successful fishing. But being part of that community doesn't really require fishing. | 999          | NA |

**Q 30 - traditional\_food**

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

| Answer          | Score Method |   |
|-----------------|--------------|---|
| In between. =3. | 3            | O |

**Q 31 - unc\_26\_30**

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

| Answer                | Score Method |    |
|-----------------------|--------------|----|
| Probably quite a lot. | 4            | NA |

**Q 32 - outdoor\_benefit**

How important is it for you to be outdoors (when you fish) for your well-being

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                           | Score Method |   |
|----------------------------------|--------------|---|
| Quite significant – give it a 4. | 4            | O |

**Q 33 - identity**

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

| Answer          | Score Method |   |
|-----------------|--------------|---|
| In between. =3. | 3            | O |

**Q 34 - home**

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

| Answer                              | Score Method |   |
|-------------------------------------|--------------|---|
| Contributes strongly, at least a 4. | 4            | O |

**Q 35 - unc\_31\_34**

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

| Answer                                                                                                                                      | Score Method |   |
|---------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|
| Doesn't vary much. The place/environment matters, of course it varies somewhat, but everyone values getting out to beautiful places.<br>=2. | 2            | O |

**Q 36 - fdays**

Approximately how many days per year do you fish?

[free text description]

| Answer                                                                                             | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit |
|----------------------------------------------------------------------------------------------------|---------------------------|--------------------------|------|
| Not in SCB's survey. Average angler maybe 5–10 days, but some 50–100 days. More than 100? Yes, HC! | 5                         | 100                      | days |

**Q 37 - income**

Approximately how large is your gross income from a fishing day on average?

[free text description]

| Answer | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit         |
|--------|---------------------------|--------------------------|--------------|
| NA     | NA                        | NA                       | not_reported |

**Q 38 - season\_maint**

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

| Answer | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit |
|--------|---------------------------|--------------------------|------|
|        |                           |                          |      |

|                                                                                                                   |     |      |            |
|-------------------------------------------------------------------------------------------------------------------|-----|------|------------|
| Servicing a boat: at least 5,000 SEK and up. Without a boat: maybe service a reel every couple of years, ~100 SEK | 100 | 5000 | sek_season |
|-------------------------------------------------------------------------------------------------------------------|-----|------|------------|

### Q 39 - fuel\_trip

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

| Answer                                                                                                  | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit       |
|---------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|------------|
| Varies hugely. Trolling anglers burn a lot. Ordinary perch & pike anglers maybe 30–50 L/trip on average | 30                        | 50                       | liter_trip |

### Q 40 - extra\_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

| Answer                                                                 | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit       |
|------------------------------------------------------------------------|---------------------------|--------------------------|------------|
| Maybe 50 L for a trolling angler. Boats are efficient, so 20–50 L/trip | 20                        | 50                       | liter_trip |

### Q 41 - extra\_crew

What are the costs for extra crew on board?

[free text description]

| Answer | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit         |
|--------|---------------------------|--------------------------|--------------|
| NA     | NA                        | NA                       | not_reported |

### Q 42 - extra\_cleaning\_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

| Answer                                                                                                                                | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit |
|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|------|
| Some make their own lures/jigs/wobblers – for them, that's half the fishing time, usually done in winter. Affected by algal bloom? No | 999                       | 999                      | hour |

### Q 43 - exytra\_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?  
[free text description]

| Answer                                                                                                                                                                                  | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|--------------|
| Yes – e.g., if you booked a salmon fishing week and they close fishing, you're stuck with costs for a cabin if it can't be rebooked. Usually you can rebook, but I don't know the cost. | NA                        | NA                       | not_reported |

### Q 44 - unc\_36\_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

| Answer | Score | Method |
|--------|-------|--------|
| 5      | 5     | O      |

### Q 45 - NA

Anything you want to add?

[free text description]

| Answer                                                                                                                                                                                                                                                   | Score    | Method |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| I think you've captured most of it. The most likely major impacts are wind – if extreme weather is prolonged, it affects access. And if you fish salmonids, a summer heatwave affects both fish and fishing opportunities. Same with winter ice fishing. | -<br>999 | NA     |

### Interviewee: I 6

6. name: lars-uno. category: pelagic trawlers id6 fish pelagic fish both in the baltic sea and the north sea. in the baltic sea, the fishing targets herring/ sprat and sharp-sprat, and the catch is sent to fishmeal. in the baltic sea, lars-uno fishes from the southern baltic up to the bothnian sea (at the level of sundsvall). they can occasionally fish in outermost regions (shetlands). the main fishing season is from august to april (but the season can extend longer into spring and summer), and the current law does not establish any limitation regarding time at sea (e.g.: seasonal ban, weekend ban). their activity is limited by the quota. in winter their activity in the baltic might be affected by ice coverage. how they adapt: pelagic trawlers have a great spatial mobility because of their large size vessels. their primary method to adapt to weather is to move around: they may switch from an area to another when they know the climatic conditions (i.e.: water temperature) are good. in the case of local weather is not good they stay in port. they have efficient cooling systems to deal with hot weather.

### Q 2 - go\_out

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1–5; from 1 – very unlikely, to 5 – very likely]

|   | Event                                                                                        | Answer                                                                                                                                                                                  | Score | Method |
|---|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season                                                                | 5                                                                                                                                                                                       | 5     | O      |
| b | Heatwave in the summer season                                                                | 1; wait a minute, spring when the weather changes; in July when it's hottest, the fish are scattered, so there's no point in going out; say 2.                                          | 1.5   | G      |
| c | Severe icing (in the winter season)                                                          | we avoid going out, 1.                                                                                                                                                                  | 1     | O      |
| d | Algal bloom (in the summer season)                                                           | 5                                                                                                                                                                                       | 5     | O      |
| e | Extreme cloudburst (in the summer season)                                                    | 5                                                                                                                                                                                       | 5     | O      |
| f | Gale                                                                                         | 4 in the Baltic Sea at least; if we are going to consume the fish in the North Sea, then we wait because the fish can spoil , but in the Baltic Sea we don't fish for human consumption | NA    | M      |
| g | Storm                                                                                        | 1, we have the quota so we don't need to risk the crew.                                                                                                                                 | 1     | O      |
| h | Are there any of these in combination that affect how likely it is that you go out and fish? | Icing & high winds, then it's 1                                                                                                                                                         | NA    | NA     |

Multiple options identified

|   | Event | area     | Score (given) |
|---|-------|----------|---------------|
| f | Gale  | Baltic   | 4             |
| f | Gale  | NorthSea | 1             |

### Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better]

|   | Event                         | Answer                                                                                                                                                                                                | Score | Method |
|---|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season | It can go both better and worse, but it has an impact; for the same weather phenomenon, it can be both better and worse; weather changes affect catchability, as do moon phases and the time of year. | 3     | NA     |

|   |                                           |                                                                                                                                                                                                                                                                          |    |   |
|---|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---|
| b | Heatwave in the summer season             | in the North Sea, herring fishing is good in the summer; in August in Skagerrakk, we saw that when the water became warm, we got large quantities of herring at Skagen when the temperature was 20-21°C; But in the Baltic Sea, the fish spread out when it gets warm. 2 | NA | M |
| c | Severe icing (in the winter season)       | We can't load the boat when it's icy. 1.                                                                                                                                                                                                                                 | 1  | O |
| d | Algal bloom (in the summer season)        | 1                                                                                                                                                                                                                                                                        | 1  | O |
| e | Extreme cloudburst (in the summer season) | Doesn't affect 3.                                                                                                                                                                                                                                                        | 3  | O |
| f | Gale                                      | 3. It can sometimes, but no 3                                                                                                                                                                                                                                            | 3  | O |
| g | Storm                                     | it can be harder for us to handle the boat and get where we want to go, so it does have an impact. 2.                                                                                                                                                                    | 2  | O |

Multiple options identified

|   | Event                         | area     | Score (given) |
|---|-------------------------------|----------|---------------|
| b | Heatwave in the summer season | Baltic   | 2             |
| b | Heatwave in the summer season | NorthSea | 4             |

#### Q 4 - catch\_assemblage

Do you catch a different composition of species during such extreme weather events?

[1-5; from 1 – very similar, to 5 – very different species composition]

|   | Event                                     | Answer | Score | Method |
|---|-------------------------------------------|--------|-------|--------|
| a | Heatwave in the winter season             | 1      | 1     | O      |
| b | Heatwave in the summer season             | NA     | 1     | NA     |
| c | Severe icing (in the winter season)       | 1      | 1     | O      |
| d | Algal bloom (in the summer season)        | 1      | 1     | O      |
| e | Extreme cloudburst (in the summer season) | 1      | 1     | O      |
| f | Gale                                      | 1      | 1     | O      |

|   |       |   |   |   |
|---|-------|---|---|---|
| g | Storm | 1 | 1 | O |
|---|-------|---|---|---|

### Q 5 - catch\_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

|   | Event                                     | Answer                                                                                                       | Score | Method |
|---|-------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | 2, because the boat moves and the fish breaks                                                                | 2     | O      |
| b | Heatwave in the summer season             | 1 same thing, in Lake Norske it becomes a problem because the boat moves and the fish breaks when it is warm | 1     | O      |
| c | Severe icing (in the winter season)       | 3                                                                                                            | 3     | O      |
| d | Algal bloom (in the summer season)        | 3                                                                                                            | 3     | O      |
| e | Extreme cloudburst (in the summer season) | 3                                                                                                            | 3     | O      |
| f | Gale                                      | 1                                                                                                            | 1     | O      |
| g | Storm                                     | 1                                                                                                            | 1     | O      |

### Q 6 - cope\_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1–5; from 1 – very unlikely, to 5 – very likely]

|   | Event                                     | Answer                                                   | Score | Method |
|---|-------------------------------------------|----------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | We cool with RSV systems, so it stays in good quality. 5 | 5     | O      |
| b | Heatwave in the summer season             | We cool with RSV systems, so it stays in good quality. 5 | 5     | O      |
| c | Severe icing (in the winter season)       | NA                                                       | -999  | NA     |
| d | Algal bloom (in the summer season)        | NA                                                       | -999  | NA     |
| e | Extreme cloudburst (in the summer season) | NA                                                       | -999  | NA     |
| f | Gale                                      | You can go ashore and unload what you have. 3            | 3     | O      |

|   |       |                                             |   |   |
|---|-------|---------------------------------------------|---|---|
| g | Storm | You can go ashore and unload what you have. | 3 | O |
|---|-------|---------------------------------------------|---|---|

### Q 7 - lag\_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

4, we saw that mackerel migrate, but the funny thing is that they are starting to come back again, as they did 20 years ago. I don't know if this has anything to do with the heat, or if it's a cycle; mackerel have been north of Shetland, but now it seems that they are back in Norway again; perhaps they will return to how it was 20-25 years ago; In the Baltic Sea, I think cod has been affected because it has gotten warmer. Thirty-five to forty years ago, the entire Baltic Sea was frozen, but sprat and herring have increased in the last five years. Smelt has also recovered because it has gotten warmer. Yes, it has affected us.

|   | Event                                     | Answer | Score | Method |
|---|-------------------------------------------|--------|-------|--------|
| a | Heatwave in the winter season             | NA     | 4     | NA     |
| b | Heatwave in the summer season             | NA     | 4     | NA     |
| c | Severe icing (in the winter season)       | NA     | -999  | NA     |
| d | Algal bloom (in the summer season)        | NA     | -999  | NA     |
| e | Extreme cloudburst (in the summer season) | NA     | -999  | NA     |
| f | Gale                                      | NA     | -999  | NA     |
| g | Storm                                     | NA     | -999  | NA     |

### Q 8 - unc\_2\_7

How much do you think the answers to these questions (questions 2-7) vary between fishers in the group you represent? 1-5; from 1 – not at all, to 5 – very much

| Answer                                                                                                        | Score | Method |
|---------------------------------------------------------------------------------------------------------------|-------|--------|
| I think they have roughly the same opinion, but there are probably some who think a little differently, so =3 | 3     | O      |

### Q 9 - personal\_safety

How much concern for your personal safety do these extreme weather events bring?

[0-2; 0 – no concern, 1 – personal injuries, 2 – life-threatening

|   | Event                         | Answer                          | Score | Method |
|---|-------------------------------|---------------------------------|-------|--------|
| a | Heatwave in the winter season | 1 if the heat causes high winds | 1     | O      |



|   |                                           |                                    |     |    |
|---|-------------------------------------------|------------------------------------|-----|----|
| b | Heatwave in the summer season             | no concern, usually calm weather   | 0   | NA |
| c | Severe icing (in the winter season)       | 2 as the boat may capsize and sink | 2   | O  |
| d | Algal bloom (in the summer season)        | 0                                  | 0   | O  |
| e | Extreme cloudburst (in the summer season) | 1                                  | 1   | O  |
| f | Gale                                      | 1                                  | 1   | O  |
| g | Storm                                     | 2, perhaps even 3                  | 2.5 | O  |

### Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

|   | Event                                     | Answer                                                                                                                                                                               | Score | Method |
|---|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | 3, [heat wave in winter, often accompanied by storms]                                                                                                                                | 3     | O      |
| b | Heatwave in the summer season             | 0                                                                                                                                                                                    | 0     | O      |
| c | Severe icing (in the winter season)       | 3 (but preferably without the trawl in, then we can get ashore)                                                                                                                      | 3     | O      |
| d | Algal bloom (in the summer season)        | 0                                                                                                                                                                                    | 0     | O      |
| e | Extreme cloudburst (in the summer season) | 0                                                                                                                                                                                    | 0     | O      |
| f | Gale                                      | 0                                                                                                                                                                                    | 0     | O      |
| g | Storm                                     | 3 can even damage the trawl; This is because we have a 40 m boat, but those with 60-80 m boats in the Baltic Sea have no problem turning with trawls during storms in the Baltic Sea | 3     | O      |

### Q 11 - minor\_damage\_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

| Answer | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit |
|--------|---------------------------|--------------------------|------|
|--------|---------------------------|--------------------------|------|

Can be repaired on board; 50,000 to repair a trawl, transport re |50000 |sek\_u  
 it 10,000 SEK; if the entire trawl were destroyed, our trawl |10000  
 costs 200,000 SEK, there are trawls that cost 800,000 SEK, if  
 you lose it Repair costs 50,000-30,000 trawls The boat is  
 rarely damaged; but a 60 m boat got a hole in the side so  
 they had to replace a plate, which cost 20-30,000 SEK, but  
 we go ash

## Q 12 - major\_damage\_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

| Answer | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit     |
|--------|---------------------------|--------------------------|----------|
| See 11 | 1e+05                     | 5e+05                    | sek_unit |

## Q 13 - change\_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0-2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

|   | Event                                     | Answer                                                                                                                                                                      | Score | Method |
|---|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | 2, stays upwind or goes ashore when the wind picks up to storm force, and waits 1-2 days until it calms down; when the waves are too high, the trawl does not fish as well. | 2     | O      |
| b | Heatwave in the summer season             | 0                                                                                                                                                                           | 0     | O      |
| c | Severe icing (in the winter season)       | 2, does not go out                                                                                                                                                          | 2     | O      |
| d | Algal bloom (in the summer season)        | 0                                                                                                                                                                           | 0     | O      |
| e | Extreme cloudburst (in the summer season) | 0                                                                                                                                                                           | 0     | O      |
| f | Gale                                      | 0                                                                                                                                                                           | 0     | O      |
| g | Storm                                     | 2, does not go out                                                                                                                                                          | 2     | O      |

## Q 14 - unc\_9\_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

| Answer                                                  | Score | Method |
|---------------------------------------------------------|-------|--------|
| Depends on the size of the boat, varies quite a lot, 4. | 4     | O      |

### Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

|   | Event                                     | Answer                                                                                                                                                | Score | Method |
|---|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | We avoid it, we can go down into the Baltic Sea on the other side of Åland; we can move, we go closer to Finland, 3-4.                                | 3.5   | O      |
| b | Heatwave in the summer season             | It has no effect                                                                                                                                      | -999  | NA     |
| c | Severe icing (in the winter season)       | Yes, we can change course if we know in advance, but if the wind is too strong, it doesn't help to move closer to land; we can't go closer than 40 km | 4     | NA     |
| d | Algal bloom (in the summer season)        | No effect                                                                                                                                             | -999  | NA     |
| e | Extreme cloudburst (in the summer season) | No effect                                                                                                                                             | -999  | NA     |
| f | Gale                                      | No effect                                                                                                                                             | -999  | NA     |
| g | Storm                                     | as we mentioned earlier                                                                                                                               | 4     | NA     |

### Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

| Answer  | Score | Method |
|---------|-------|--------|
| Yes, 2. | 2     | NA     |

### Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer                                                                                                                                                                                     | Score | Method |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| We have a few ports where we can land, but there's a small problem at the moment because there have been some ideological issues in Norrsundet, so now we're going to Oxelésund instead. 1 | 1     | NA     |

**Q 18 - management\_space**

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer                                                                                      | Score Method |   |
|---------------------------------------------------------------------------------------------|--------------|---|
| Yes, I can choose the North Sea or Skagerrak instead if it's too windy in the Baltic Sea. 2 | 2            | O |

**Q 19 - management\_species**

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer    | Score Method |    |
|-----------|--------------|----|
| Not asked | -            | NA |
|           | 999          |    |

**Q 20 - unc\_15\_19**

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

| Answer                                 | Score Method |   |
|----------------------------------------|--------------|---|
| I think they think the same (=1 or 2?) | 1.5          | O |

**Q 21 - stress**

How much stress does the risk of the following events cause (in general)?

[1–5; from 1 – really not stressful, to 5 – very stressful]

|   | Type             | Answer                                                                                                                  | Score Method |    |
|---|------------------|-------------------------------------------------------------------------------------------------------------------------|--------------|----|
| a | stress_injuries  | Quite high, it's always there lurking, you're always a little worried, 4, because my 15-year-old son has to go with us. | 4            | O  |
| b | stress_gear      | Also stressful                                                                                                          | 4            | NA |
| c | stress_condition | yes, it's also stressful because we lose income, quite high stress = 4                                                  | 4            | O  |

**Q 22 - unc\_21**

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

| Answer                         | Score Method |   |
|--------------------------------|--------------|---|
| I think it's pretty similar, 2 | 2            | O |

### Q 23 - job\_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

| Answer                                                              | Score Method |   |
|---------------------------------------------------------------------|--------------|---|
| no; some have tried to engage in wind power, with smaller boats. =1 | 1            | O |

### Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

| Answer                                                                                                                                                                                                                                                                          | Score Method |   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|
| Yes, we are working on that, herring burgers, producing more products from the Baltic Sea for human consumption; we have to adapt, if it is to be more humane we have to adjust and go ashore after maybe 1-2 days; now we are out for a week; to maintain quality. (=4 or 5?). | 4.5          | O |

### Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1–5; from 1 – very difficult, to 5 – very easy]

| Answer                                                                                                                        | Score Method |   |
|-------------------------------------------------------------------------------------------------------------------------------|--------------|---|
| Loans are no problem, we got help developing the equipment, it's much worse today; but you can get help to buy a new trawl =5 | 5            | O |

### Q 26 - unc\_23\_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

| Answer                                     | Score Method |   |
|--------------------------------------------|--------------|---|
| large variation between different boats, 5 | 5            | O |

**Q 27 - knowledge\_transmission**

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

| Answer             | Score Method |   |
|--------------------|--------------|---|
| very important, 5. | 5            | O |

**Q 28 - benefit\_local\_community**

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                                                                                     | Score Method |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|
| we do a little repair work there; we don't land any fish in Fiskebäck; it could be better if we landed there; to a certain extent, but somewhat average, 3 | 3            | O |

**Q 29 - benefit\_personal\_community**

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                                                                        | Score Method |   |
|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|
| For my colleagues? Well, you want to have your colleagues, because there are so few of us now, so you want to keep all your colleagues.<br>=5 | 5            | O |

**Q 30 - traditional\_food**

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                   | Score Method |   |
|------------------------------------------------------------------------------------------|--------------|---|
| I think it's very important, when we consume herring, canned herring, it's important, =5 | 5            | O |

**Q 31 - unc\_26\_30**

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

| Answer                                                                                                                                                                                   | Score Method |    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----|
| It varies, because we are pelagic, and then you have the demersal fishermen, who are a different species; among the pelagic fishermen, it is quite similar, it is important to everyone. | 2            | NA |

**Q 32 - outdoor\_benefit**

How important is it for you to be outdoors (when you fish) for your well-being

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                    | Score Method |    |
|---------------------------------------------------------------------------|--------------|----|
| It's good to get out on deck, get out on the sea? It's an important part! | 4            | NA |

**Q 33 - identity**

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

| Answer      | Score Method |    |
|-------------|--------------|----|
| it's strong | 5            | NA |

**Q 34 - home**

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

| Answer                                                           | Score Method |    |
|------------------------------------------------------------------|--------------|----|
| Yes, Skagen is a really fantastic place, actually very important | 5            | NA |

**Q 35 - unc\_31\_34**

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

| Answer                                       | Score Method |    |
|----------------------------------------------|--------------|----|
| I think everyone thinks pretty much the same | 2            | NA |

**Q 36 - fdays**

Approximately how many days per year do you fish?

[free text description]

| Answer                                                                                                                                                                                         | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|------|
| In the Baltic Sea, from November to June, about half a year, 7 months, at the moment, but that can change; we fish about 80% of those days, are on board and have the trawl in and are fishing | 140                       | 170                      | days |

### Q 37 - income

Approximately how large is your gross income from a fishing day on average?

[free text description]

| Answer                                                                                                                                                                                                                                                                                                                       | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|--------------------|
| It varies; if we're talking about all boats, we collaborate with another boat, and we've exchanged some fish between us; in the Bothnian Sea, [calculating] 11,000/hour, 25,000 per day in the Bothnian Sea; but off Shetland, we caught 400 tons of mackerel, the entire annual quota, in one hour! That was 8 million/hour | 25                        | 30                       | Million_sek_season |

### Q 38 - season\_maint

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

| Answer                                                                                                                                                                                                                                                                                                                                                                                              | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|-------------|
| We are doing extensive renovations now, because we are doing a 10-year class when we go through the entire boat, so it could be 10% of the catch, in a normal year it is 1% of the catch (has an income of SEK 25-30 million per year), so 1% of that. If there are two trawls in a season, SEK 300-400. We have an extra trawl with us, as we have two drums, and we have 1-2 trawls in Norrsundet | 1                         | 10                       | % of income |

### Q 39 - fuel\_trip

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

| Answer                                                                                                                                                                                                                                    | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|-------------|
| if a fishing trip lasts about 1 week; in the Bothnian Sea; 50 m3/day * 6 days* = 240 kkr/week 2.5 tons * 4.5*22 hours; runs 22 hours per day 5000L*8h/day; for one week 240 kkr/week (in the Bothnian Sea) [talks about mackerel fishing] | 600                       | 650                      | liters_hour |

### Q 40 - extra\_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]



| Answer                                                                                                                                                  | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|--------------|
| transport by truck to Skagen from Norrsundet; do not want to drive the boat that far, it costs money; the Bothnian Sea to the Baltic Sea takes 1.5 days | NA                        | NA                       | not_reported |

#### Q 41 - extra\_crew

What are the costs for extra crew on board?

[free text description]

| Answer                                                                                                                                                                                                     | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|--------------|
| Occasionally, if someone is injured or unavailable; there are four of us on the boat; previously, when we had the net, we needed nine people, but that was ten years ago; substitutes receive the same pay | NA                        | NA                       | not_reported |

#### Q 42 - extra\_cleaning\_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

| Answer                                                                                                                                                                      | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|-------|
| No, we don't need to clean the trawls; we lower them into the sea upside down so that they get clean; we both hose them down and dip them; no difference with algae blooms. | 0                         | 0                        | hours |

#### Q 43 - exytra\_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?

[free text description]

| Answer                                                                                                                                                                                                                                                                                | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|--------------|
| Yes, a window can break if you hit a big wave. One boat had a compass on the roof that broke and made a hole right through the roof, so it can cause damage and break things. If water gets into the boat and damages the electronics, it becomes more expensive. 10,000 for a window | NA                        | NA                       | not_reported |

#### Q 44 - unc\_36\_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

| Answer                                                                                                                                                                                                                                                                                                                                                                   | Score | Method |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| It varies a lot, many have a very new boat, so the costs are very low, and if you have a larger boat, it can withstand extreme weather better. Astrid loads > 2000 tons, while we load 500 tons, and we have to make 4 trips back and forth, while they only make 1 trip, so they probably have lower fuel costs per fish, even though they consume more when they drive | 5     | NA     |

#### Q 45 - NA

Anything you want to add?

[free text description]

| Answer                                                                                                                                                                                                     | Score    | Method |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| In terms of extreme weather with storms, I think we've covered everything; this large variation in temperature in the Baltic Sea is something you're looking at, I think, with temperature data from SMHI. | -<br>999 | NA     |

#### Interviewee: I 7

7. name: lars ivan h  llstrand. category: coastal fisherman id7 is a professional coastal fisherman in the southwestern bothnian sea (  regrund area?). they fish with gillnets and traps. the main fishing season is summer. they can do some ice fishing. the main target is herring, whitefish, salmon, perch. they have more salmon traps than the average coastal fisherman. fishing for perch (nets) is done from a small boat close to home, while fishing for herring happens with a larger boat more far away. a trip for herring can cost up to 100 litres of diesel (approximately 10 l\*hour at 10 knots means that they can go to cast at least 50-60 km from home). how they adapt: they primarily adapt when fishing with nets to hw by changing fishing target. when it's hot they aim for perch and not for herring. in addition, they can perform shorter sets and use cooling boxes. can also adapt to hw and storms by changing fishing location. they are not affected by ab. to avoid hw they can go further out in deeper waters, the opposite for storms. in any case they can move on short distances. their spatial mobility is mainly affected by the boat and infrastructure (they want to go back home) on second instance by management that makes possible to fish in proximity. there is not the same flexibility for salmon traps, which are fixed.

#### Q 2 - go\_out

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1-5; from 1 – very unlikely, to 5 – very likely]

|   | Event                               | Answer                                                                                                            | Score | Method |
|---|-------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season       | I probably go fishing anyway, it's the wind that matters, I probably try anyway (=5? or =4?)                      | 4.5   | O      |
| b | Heatwave in the summer season       | I don't go because the fish won't keep as well, I don't fish for herring at all, in that case I fish for perch, 2 | 2     | O      |
| c | Severe icing (in the winter season) | I am not on the lake at that time of year when there is a risk of icing (because the ice forms so early           | -999  | NA     |

|   |                                                                                              |                                                                                                                                                                                                                                                                                                               |    |    |
|---|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|
| d | Algal bloom (in the summer season)                                                           | I have not experienced it, but it probably has an impact, so then I don't fish                                                                                                                                                                                                                                | 1  | NA |
| e | Extreme cloudburst (in the summer season)                                                    | then I fish anyway                                                                                                                                                                                                                                                                                            | 4  | NA |
| f | Gale                                                                                         | depends on the wind direction, if it's N then it's from the outside so then it's difficult, if it's S or SW, in gale force winds I don't fish anyway because there's a strong current in the lake so you get crap in your nets, but I do go out in gale force winds sometimes , depends on the wind direction | 3  | NA |
| g | Storm                                                                                        | 1                                                                                                                                                                                                                                                                                                             | 1  | O  |
| h | Are there any of these in combination that affect how likely it is that you go out and fish? | storms and weather like that                                                                                                                                                                                                                                                                                  | NA | NA |

Multiple options identified

|   | Event                         | target  | Score (given) |
|---|-------------------------------|---------|---------------|
| b | Heatwave in the summer season | herring | 1             |
| b | Heatwave in the summer season | perch   | 2             |

### Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1-5; from 1 – much worse, 3 – no difference, to 5 – much better]

|   | Event                                     | Answer                                                                                                                                                                              | Score | Method |
|---|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | difficult to say, haven't experienced it that much, am more limited when it's windy because seaweed gets caught in the nets (enough if it's gale force), f. fishing gets much worse | -999  | NA     |
| b | Heatwave in the summer season             | not when it comes to perch, but herring is much more sensitive to heat, so it gets much worse                                                                                       | NA    | NA     |
| c | Severe icing (in the winter season)       | We don't ask about that                                                                                                                                                             | -999  | NA     |
| d | Algal bloom (in the summer season)        | No, I've only heard that fishing gets worse; I've seen some green filamentous algae on the surface this year (I've seen a little in previous years; it never stays on the surface)  | 2     | NA     |
| e | Extreme cloudburst (in the summer season) | 3                                                                                                                                                                                   | 3     | O      |
| f | Gale                                      | 1                                                                                                                                                                                   | 1     | O      |

|   |       |   |   |   |
|---|-------|---|---|---|
| g | Storm | 1 | 1 | O |
|---|-------|---|---|---|

Multiple options identified

|   | Event                         | target  | Score (given) |
|---|-------------------------------|---------|---------------|
| b | Heatwave in the summer season | herring | 3             |
| b | Heatwave in the summer season | perch   | 1             |

#### Q 4 - catch\_assemblage

Do you catch a different composition of species during such extreme weather events?

[1–5; from 1 – very similar, to 5 – very different species composition]

|   | Event                                     | Answer                                                                                                                                          | Score | Method |
|---|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | difficult to answer                                                                                                                             | -999  | NA     |
| b | Heatwave in the summer season             | herring are fished a little further out, and perch closer in; always fish in different places, so the composition is not affected in each place | 2     | NA     |
| c | Severe icing (in the winter season)       | We skip this one                                                                                                                                | -999  | NA     |
| d | Algal bloom (in the summer season)        | Haven't experienced it, but heard that it affects fishing                                                                                       | -999  | NA     |
| e | Extreme cloudburst (in the summer season) | 1                                                                                                                                               | 1     | O      |
| f | Gale                                      | Yes, I think it affects the composition; seaweed gets caught in the nets and creates currents; can't say if other species are affected          | -998  | NA     |
| g | Storm                                     | Yes, I think it affects the composition; seaweed gets caught in the nets and creates currents; can't say if other species are affected          | -998  | NA     |

#### Q 5 - catch\_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

|   | Event                         | Answer | Score | Method |
|---|-------------------------------|--------|-------|--------|
| a | Heatwave in the winter season | 3      | 3     | O      |

|   |                                           |                                                                                                                                                                                             |      |    |
|---|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----|
| b | Heatwave in the summer season             | Yes, the warmer it is, the shorter the shelf life of the fish, especially if it is fatty fish (herring, salmon, and whitefish), which is in the summer; herring is particularly sensitive 1 | 2    | M  |
| c | Severe icing (in the winter season)       | Not asked                                                                                                                                                                                   | -999 | NA |
| d | Algal bloom (in the summer season)        | Cannot answer                                                                                                                                                                               | -999 | NA |
| e | Extreme cloudburst (in the summer season) | 3                                                                                                                                                                                           | 3    | O  |
| f | Gale                                      | No, I don't think it's the quality, it's probably the catch volume 3                                                                                                                        | 3    | O  |
| g | Storm                                     | 3                                                                                                                                                                                           | 3    | O  |

Multiple options identified

|   | Event                         | target           | Score (given) |
|---|-------------------------------|------------------|---------------|
| b | Heatwave in the summer season | salmon_whitefish | 1.5           |
| b | Heatwave in the summer season | herring          | 1             |

#### Q 6 - cope\_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1–5; from 1 – very unlikely, to 5 – very likely]

|   | Event                                     | Answer                                                                                                                                | Score | Method |
|---|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | NA                                                                                                                                    | -999  | NA     |
| b | Heatwave in the summer season             | Yes, a little (3), I can fish more often, for perch you can keep the nets in for shorter periods, the fish freeze as soon as I get in | 3     | O      |
| c | Severe icing (in the winter season)       | NA                                                                                                                                    | -999  | NA     |
| d | Algal bloom (in the summer season)        | NA                                                                                                                                    | -999  | NA     |
| e | Extreme cloudburst (in the summer season) | NA                                                                                                                                    | -999  | NA     |
| f | Gale                                      | NA                                                                                                                                    | -999  | NA     |
| g | Storm                                     | NA                                                                                                                                    | -999  | NA     |

### Q 7 - lag\_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

Of course it has changed, but we don't know why; more storms and such have a certain impact on how I fish, 3-4

|   | Event                                     | Answer | Score | Method |
|---|-------------------------------------------|--------|-------|--------|
| a | Heatwave in the winter season             | NA     | -999  | NA     |
| b | Heatwave in the summer season             | NA     | -999  | NA     |
| c | Severe icing (in the winter season)       | NA     | -999  | NA     |
| d | Algal bloom (in the summer season)        | NA     | -999  | NA     |
| e | Extreme cloudburst (in the summer season) | NA     | -999  | NA     |
| f | Gale                                      | NA     | 3.5   | NA     |
| g | Storm                                     | NA     | 3.5   | NA     |

### Q 8 - unc\_2\_7

How much do you think the answers to these questions (questions 2-7) vary between fishers in the group you represent? 1-5; from 1 – not at all, to 5 – very much

| Answer                                           | Score | Method |
|--------------------------------------------------|-------|--------|
| roughly the same, roughly the same conditions, 2 | 2     | O      |

### Q 9 - personal\_safety

How much concern for your personal safety do these extreme weather events bring?

[0-2; 0 – no concern, 1 – personal injuries, 2 – life-threatening

|   | Event                               | Answer                                                                                                                                                                      | Score | Method |
|---|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season       | if the ice becomes unsafe, there is a risk of falling into the water, which can be life-threatening; if you have equipment in the lake, you want to be able to access it, 2 | 2     | O      |
| b | Heatwave in the summer season       | 0                                                                                                                                                                           | 0     | O      |
| c | Severe icing (in the winter season) | not asked                                                                                                                                                                   | -999  | NA     |

|   |                                           |   |   |   |
|---|-------------------------------------------|---|---|---|
| d | Algal bloom (in the summer season)        | 0 | 0 | O |
| e | Extreme cloudburst (in the summer season) | 0 | 0 | O |
| f | Gale                                      | 2 | 2 | O |
| g | Storm                                     | 2 | 2 | O |

### Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

|   | Event                                     | Answer                                                                                                           | Score | Method |
|---|-------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | 0                                                                                                                | 0     | O      |
| b | Heatwave in the summer season             | 0                                                                                                                | 0     | O      |
| c | Severe icing (in the winter season)       | not asked                                                                                                        | -999  | NA     |
| d | Algal bloom (in the summer season)        | Yes, it could be on the gear in the nets, getting clogged, so 1                                                  | 1     | O      |
| e | Extreme cloudburst (in the summer season) | 0 , the boat could fill with water, but that would have to be extreme; if you have an open boat, it could happen | 1     | G      |
| f | Gale                                      | On gear and on the boat; 1                                                                                       | 1     | O      |
| g | Storm                                     | . 2, gear could be completely destroyed, such as salmon traps and so on                                          | 2     | O      |

### Q 11 - minor\_damage\_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

| Answer                                                                                                                                                                                                                                                                                                                                                                                                                       | Score - low (given) | Score - up (given) | Unit     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------|----------|
| You know about nets, that's minor damage, but salmon traps cost more to repair because they're special equipment; a few thousand for nets, but salmon traps can cost tens of thousands to repair. Yes, on the boat, it's probably if you drive in bad weather, and there is greater damage to the propeller and gearbox on smaller boats, and on large boats it's also the propeller, then it's tens of thousands of kronor. | 1000                | 50000              | sek_unit |

**Q 12 - major\_damage\_cost**

What are the costs of repairing major damages to gear and boat?

[free text description]

| Answer | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit     |
|--------|---------------------------|--------------------------|----------|
| see 11 | 50000                     | 100000                   | sek_unit |

**Q 13 - change\_practices**

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0–2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

|   | Event                                     | Answer                                                                                                                                                 | Score | Method |
|---|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | Yes, if there is no ice, I cannot fish, so I have to fish from a boat, but that is usually not very productive; 2                                      | 2     | O      |
| b | Heatwave in the summer season             | 1 changes how much perch I fish compared to herring                                                                                                    | 1     | O      |
| c | Severe icing (in the winter season)       | I don't know, I haven't experienced it (maybe I'll skip it then),                                                                                      | -999  | NA     |
| d | Algal bloom (in the summer season)        | NA                                                                                                                                                     | -999  | NA     |
| e | Extreme cloudburst (in the summer season) | 0                                                                                                                                                      | 0     | O      |
| f | Gale                                      | Changes the fishing spot                                                                                                                               | 1     | NA     |
| g | Storm                                     | 2; Do you go out in a storm? It has happened that you make a mistake and go out, but you have to put the nets in a sheltered spot, you have your spots | 2     | O      |

**Q 14 - unc\_9\_13**

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

| Answer                                                                             | Score | Method |
|------------------------------------------------------------------------------------|-------|--------|
| They probably have similar conditions, so they probably answer quite similarly; =2 | 2     | O      |

**Q 15 - avoidance**

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]



|   | Event                                     | Answer                                                                                                                                                                          | Score | Method |
|---|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| a | Heatwave in the winter season             | 1 (if the ice is not strong enough, it's pretty much the same, then you have to go very far),                                                                                   | 1     | O      |
| b | Heatwave in the summer season             | I don't know, but you can still fish for perch; the roach moves further out, and I can go a little further out, but I don't have a very big boat, so I can't go that far out, 2 | 2     | O      |
| c | Severe icing (in the winter season)       | I think so , it depends on how it accumulates in certain places                                                                                                                 | 4     | NA     |
| d | Algal bloom (in the summer season)        | NA                                                                                                                                                                              | -999  | NA     |
| e | Extreme cloudburst (in the summer season) | 1 (but it doesn't matter)                                                                                                                                                       | 1     | O      |
| f | Gale                                      | 4, depends on the wind direction, and sometimes the fishing spots don't match where it's sheltered.                                                                             | 4     | O      |
| g | Storm                                     | same there, but you rarely fish in a storm if you know about it, but sometimes you have to go out as I said                                                                     | 4     | NA     |

### Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0-2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

| Answer                                                                                                                                        | Score (self) | Score (given) |
|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------|
| I have mobile gear (perch can move distance =1), herring =1 (somewhat limited by boat), but I cannot choose the location for fixed gear (= 0) | M            | NA            |

Multiple options identified

| target        | Score (given) |
|---------------|---------------|
| not_specified | 1             |
| salmon        | 0             |

### Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0-2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer | Score | Method |
|--------|-------|--------|
|--------|-------|--------|

Extremely, there are other places to go ashore; fixed with the salmon 0 O  
because then you have to report the landing port before you unload  
them (and when you unload); for the herring, I have the shaking  
machine in the boathouse, so it is difficult to find out the catch  
elsewhere, but I can go to an emergency port, i.e. =0

---

### Q 18 - management\_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer                                                                                                                                                                                                  | Score | Method |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| yes, in a limited area; the license determines where you can fish; we have quotas and fixed gear in fixed locations for salmon, but with mobile gear you can move around in the local area at least; =1 | 1     | O      |

### Q 19 - management\_species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

| Answer                                                                                                                                                                                                                                            | Score | Method |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| There isn't much else to fish for; we catch other fish species, but in much smaller quantities, such as pike and pikeperch; salmon is subject to quotas; herring too, but we don't fill the coastal quota, and haven't done so for several years. | 999   | NA     |

### Q 20 - unc\_15\_19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

| Answer                                                                | Score | Method |
|-----------------------------------------------------------------------|-------|--------|
| Not everyone has salmon traps, but otherwise it's pretty similar; =2. | 2     | G      |

### Q 21 - stress

How much stress does the risk of the following events cause (in general)?

[1–5; from 1 – really not stressful, to 5 – very stressful]

| Type              | Answer                            | Score | Method |
|-------------------|-----------------------------------|-------|--------|
| a stress_injuries | 1 no, I don't think about it much | 1     | O      |

|   |                  |                                                                                                                                                                                                                                               |   |   |
|---|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
| b | stress_gear      | sometimes, when the weather is bad, windy, stormy, that's when you feel like you have stuff out there, it's mostly the salmon traps you think about; you usually have time to remove the movable gear if you know bad weather is coming;<br>4 | 4 | O |
| c | stress_condition | I never think about it, 1                                                                                                                                                                                                                     | 1 | O |

### Q 22 - unc\_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

| Answer                                       | Score | Method |
|----------------------------------------------|-------|--------|
| I would think they would answer the same, =1 | 1     | O      |

### Q 23 - job\_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1-5; from 1 – very unlikely, to 5 – very likely]

| Answer      | Score | Method |
|-------------|-------|--------|
| a little; 3 | 3     | O      |

### Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1-5; from 1 – very unlikely, to 5 – very likely]

| Answer                                                                                                                               | Score | Method |
|--------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| I try to develop fishing so that it is as easy as possible to do certain things, and I try a little bit to find new ways to sell. =4 | 4     | O      |

### Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1-5; from 1 – very difficult, to 5 – very easy]

| Answer                                                                                                                                                                                                                                   | Score | Method |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| It probably depends on how much money and what for. You need to have good ideas. So, to buy a salmon trap? It's a bit uncertain because salmon stocks are low at the moment. A filleting machine? It's probably not entirely impossible. | 2     | NA     |

**Q 26 - unc\_23\_25**

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

| Answer                                                                                   | Score Method |   |
|------------------------------------------------------------------------------------------|--------------|---|
| I think it would vary a bit; some don't do much processing and mostly want to fish, so 4 | 4            | O |

**Q 27 - knowledge\_transmission**

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                                                                                                                                                                       | Score Method |   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|
| I've given up. I wish it could continue, but it's so difficult, there are so many problems, with the authorities, among other things. Yes, I think it's important. I'm interested, but the financial aspect is difficult for a new start. =5 | 5            | O |

**Q 28 - benefit\_local\_community**

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                                                                   | Score Method |   |
|------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|
| They get fresh fish nearby; quite a few people shop there, people come to his home to buy fish, and not just when he's at the market; =5 | 5            | O |

**Q 29 - benefit\_personal\_community**

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                                                                                                                                                                                                                                                                                                       | Score Method |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|
| No, there are so few who fish. Do you share experiences? Yes, we benefit from each other, in the past they couldn't talk to each other. Sometimes it was a matter of life and death, you've heard about that, it was when there were several in each village, but now there are so few of us that it's easy to get along. =4 | 4            | O |

**Q 30 - traditional\_food**

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

| Answer               | Score | Method |
|----------------------|-------|--------|
| 5, that's what I do. | 5     | O      |

### Q 31 - unc\_26\_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

| Answer                                   | Score | Method |
|------------------------------------------|-------|--------|
| Some, and some would say differently, =3 | 3     | O      |

### Q 32 - outdoor\_benefit

How important is it for you to be outdoors (when you fish) for your well-being  
[1–5; from 1 – not important at all, to 5 – very important]

| Answer                                              | Score | Method |
|-----------------------------------------------------|-------|--------|
| yes, it is quite important, it is very important =5 | 5     | O      |

### Q 33 - identity

How much does fishing contribute to your identity and self-image  
[1–5; from 1 – very little, to 5 – very much]

| Answer                                                    | Score | Method |
|-----------------------------------------------------------|-------|--------|
| quite a lot (=4 or 5? [He doesn't use any superlatives!]) | 4.5   | O      |

### Q 34 - home

How much does fishing contribute to creating a bond to local places  
[1–5; from 1 – very little, to 5 – very much]

| Answer                                                                                                                                    | Score | Method |
|-------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| yes, but I usually go to the same place, and stupidly enough, I don't try out many different places; some places are special to visit =4? | 4     | O      |

### Q 35 - unc\_31\_34

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

| Answer                              | Score | Method |
|-------------------------------------|-------|--------|
| They would probably be similar; =2. | 2     | O      |

**Q 36 - fdays**

Approximately how many days per year do you fish?

[free text description]

| Answer          | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit |
|-----------------|---------------------------|--------------------------|------|
| around 200 days | NA                        | 200                      | days |

**Q 37 - income**

Approximately how large is your gross income from a fishing day on average?

[free text description]

| Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|------|
| It depends on processing, fishing, and smoking; it varies at different times of the year. Spring and early summer are quite intense, then it slows down. You don't get your money back for surströmming until you've sold the cans. Varies from SEK 10,000-50,000/day, depending on the catch and salmon and such; then it is processed. If you get 100 salmon in a trap in one day, that's worth quite a lot just raw, then you smoke it and save it, so you can spread it out over the | year<br>  1000            | 50000                    | se   |

**Q 38 - season\_maint**

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

| Answer                                                                                                                                                                                     | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|------------|
| Nets require quite a lot of work, and salmon traps in the spring; say SEK 10,000-35,000/year. And in terms of time? About 30 days, depending on whether you encounter breakdowns and such. | 10000                     | 35000                    | sek_season |

**Q 39 - fuel\_trip**

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

| Answer | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit |
|--------|---------------------------|--------------------------|------|
|        |                           |                          |      |

|                                                                                                                                                                          |   |    |            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|------------|
| 30-50 liters of diesel for herring; if fishing for perch, the costs are not so high because it is quite close, then 4-5 liters of gasoline (because it is a small boat). | 5 | 50 | liter_trip |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|------------|

#### Q 40 - extra\_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

| Answer                     | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit       |
|----------------------------|---------------------------|--------------------------|------------|
| 100L of diesel for herring | 50                        | 100                      | liter_trip |

#### Q 41 - extra\_crew

What are the costs for extra crew on board?

[free text description]

| Answer                                                                                                   | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit     |
|----------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|----------|
| has one specifically for herring fishing; he does not fish that many days; costs SEK 60,000-100,000/year | 60000                     | 100000                   | sek_year |

#### Q 42 - extra\_cleaning\_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

| Answer                                                                                                                                                                                                                                                                                                                                                                    | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|-------|
| Rinses the salmon traps, takes a few hours in both spring and fall, approx. 20 hours; extra work if there is algae blooms; sometimes there is growth and sometimes there is moss floating around, which is difficult to rinse off, so it has to lie there and rot first; it grows on what is on the beaches and if it is windy, it comes loose and gets stuck in the trap | 20                        | NA                       | hours |

#### Q 43 - exytra\_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?

[free text description]

| Answer                                               | Score<br>- low<br>(given) | Score<br>- up<br>(given) | Unit         |
|------------------------------------------------------|---------------------------|--------------------------|--------------|
| No, it's usually repairs if there is any damage, etc | NA                        | NA                       | not_reported |

#### Q 44 - unc\_36\_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

| Answer             | Score Method |   |
|--------------------|--------------|---|
| Yes, I think so, 5 | 5            | O |

#### Q 45 - NA

Anything you want to add?

[free text description]

| Answer                                                                                                                                                   | Score Method |    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----|
| I don't have much experience with it yet, but what we've seen is milder winters and more wind; things have changed, but we don't know what the cause is. | -<br>999     | NA |

Pending interpretation